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UNIVERSITÄT
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Hybrid Transcriptome Assembly: Review, Benchmark and Application to the Transcriptome of the Cave Salamander

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Submitted by:

Simon Hegele

Born 11.10.1996 in Mannheim-Neckarau

Supervised by:

Prof. Dr. Arne Sahm

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Abstract

English

Newly developed hybrid approaches to hybrid *de novo* transcriptome assembly allow the use of complementary data: the short and accurate reads from Illumina's massively parallel sequencing on the one hand, and the long and error-prone reads from ONT or PacBio's single molecule real-time sequencing on the other hand. These methods, however, have not yet been systematically and comprehensively investigated. In this paper, I provide an overview of the methods of hybrid transcriptome assembly and compare them using real Illumina and matching simulated ONT data for the model organism *mus musculus*. Transcriptomes assembled from uncorrected ONT reads using RNA-Bloom surpassed all others in terms of completeness and structural correctness. Subsequent hybrid polishing of the transcript sequences using Illumina reads also significantly reduced the number of base errors. These findings were then applied to assemble the transcriptome of the olm (*Proteus anguinus*), an organism that could be of particular importance for ageing research due to its regenerative abilities and exceptional longevity.

German

Neu entwickelte hybride Ansätze zur hybriden *de novo* Transkriptomassemblierung erlauben die Nutzung komplementärer Daten: Auf der einen Seite die kurzen und akkuraten Reads der massiv parallelen Sequenzierung von Illumina und auf der anderen Seite die langen und stärker fehlerbehafteten Reads der Single Molecule Real-time Sequenzierung von ONT bzw. PacBio. Bislang wurden diese Methoden allerdings nicht systematisch und umfassend untersucht. In dieser Arbeit gebe ich eine Übersicht zu den Methoden der hybriden Transkriptomassemblierung und vergleiche diese anhand realen Illumina- und passenden simulierten ONT-Daten für den Modellorganismus *mus musculus*. Transkriptome, die von RNA-Bloom aus unkorrigierten ONT-Reads assembliert wurden, übertrafen alle anderen in der Vollständigkeit und der strukturellen Korrektheit. Durch ein anschließendes hybrides Polishing der Transkriptsequenzen mithilfe der Illumina-Reads konnten darüber hinaus die Anzahl der Basenfehler deutlich reduziert werden. Die gewonnen Erkenntnisse wurden dann zur Assemblierung des Transkriptoms des Höhlensalamanders (*Proteus anguinus*) angewendet, einem Organismus, der aufgrund seiner Regenerationsfähigkeiten und außergewöhnlichen Langlebigkeit für die Alterungsforschung von besonderer Bedeutung sein könnte.

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1. Introduction

1.1. Background & motivation

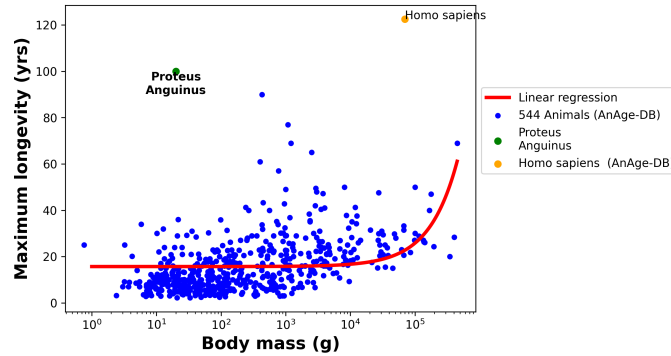


Figure 1.1: **Relationship between body mass and maximum longevity.**

For the records 544 animals (blue dots) and the human (orange dot) from the AnAge database [1] and the *Proteus Anguinus* their respective body mass is plotted against their maximum longevity. The linear regression model (red line) reflects the positive correlation between those two variables. The plot highlights the exceptionally long live span of the olm.

The olm (*Proteus Anguinus*) is an amphibian endemic to the Dinaric carst. Although it lives in cave systems, it is exposed to allogenic waters and therefore changing environmental conditions, to which it has adapted well. For example the olm can tolerate medium light exposure, a variety of tempratures and of water hardness levels. [2] Furthermore, the olm has developed behavioral and metabolic strategies that allow it to withstand prolonged periods of food deprivation. [3, 4]

Despite their small body weight of only around 20g their maximum predicted lifespan of more than 100 years is exeptionally high as illustrated in fig. 1.1. This makes the olm a promising alternative model for age related research. Two of the most common mechnisms for extending lifespans are reduced basal metabolic rates and increased antioxidant activities. However, the olms oxygen consumption, which is often used to measure the metabolic rate, does not differ substantially from those of amphibians of similar size. [5] The cave salamander is highly tolerant to anoxia and does not display typical signs of oxidative stress during reoxygenation. A common response to anoxia is an underexpression of antioxidant enzymes (e.g. superoxide dismutase, catalase and glutathione peroxidase) during anoxia itself and an overexpression thereof during reoxygenation. However, these are only partially present in the olm. It seems to pursue other, less common but effective strategies for combating reactive oxidative species. [6]

Decoding the genomic sequence of the olm may provide insight into the molecular mechanisms that

enable the *Proteus anguinus* to outlive all of its relatives. These are potentially also of interest for human biomedical research. The international Proteus Genome Research Consortium has taken on this challenging task, as the genomes enormous size of an estimated 50 gigabases poses a major difficulty. [7] So far, they did, however, not report any progress. When genomes are inaccessible, *de novo* transcriptome assembly offers a suitable alternative, serving as proxy-catalogue of expressed genes.[8]. A first, but scarce *de novo* assembled transcriptome for the olm was published by Irrisari et al., 2017. [9]. Earlier this year, we have published a more complete assembly using Illumina short reads and ONT nanopore long reads. [10]

Newly developed hybrid approaches to transcriptome assembly allow the integrated use of complementary data from Illumina and PacBio (Pacific Biosciences) or ONT (Oxford Nanopore Technologies). In the massively parallel sequencing by synthesis, large quantities of DNA fragments are distributed and anchored on a flow cell, where they are amplified locally by a process known as bridge amplification. This results in clusters of identical single-stranded DNA fragments. The complementary strand is then synthesised step by step through the iterative insertion of individual fluorescent nucleotide bases. The synthesised DNA sequence can be read using light signals from the inserted fluorescent nucleotide bases. Since large numbers of identical strands are synthesised in parallel, failures in the insertion of bases in individual strands are less significant. Consequently, Illumina reads generally achieve high accuracy. At the same time, however, their length is very limited. Firstly, because longer DNA fragments would result in a blending of different fragment clusters. Secondly, synthesis errors accumulate during the sequences and which leads to an increasing decline in read quality. The length of individual Illumina reads is usually less than 300 bases. Paired-end sequencing allows DNA fragments, which are also only a few hundred bases long, to be read from both ends. With PacBio and ONT, on the other hand, tens of thousands of bases can be read in at a time. PacBio is also based on the sequencing-by-synthesis principle, but for individual molecules. In ONT sequencing, an electrical voltage is applied to a membrane. An electrical current flowing through nanopores in this membrane naturally attempts to resolve this voltage. A motor protein guides individual DNA or RNA strands through this pore, with the nucleotide bases hindering the flow of electrons to varying degrees due to their different structures. Both technologies show significantly lower accuracy compared to Illumina sequencing. Hybrid approaches attempt to combine the advantages of Illumina (their accuracy) with those of PacBio or ONT (their continuity). Hybrid approaches attempt to improve *de novo* transcriptome assembly by leveraging the respective advantages of both types of reads. However, these methods have not yet been systematically and comprehensively investigated. Multiple comparative benchmarks that extensively tested software tools for hybrid long read error correction that utilize complementary short reads to reduce error rates in long read data sets have been performed in the context of genomic long read correction. [11, 12, 13] Lima et al. (2020) have performed a thorough analysis of long read error correction tools applied to transcriptomic reads. [14] However, they used RNA sequencing data of only a single tissue, when, due to the high costs of sequencing in practice the RNA of multiple tissues are often pooled and sequenced together. Furthermore, the study does not include more recently published and expression-level aware software tools that have been specifically designed for the correction of transcriptomic sequence reads.

The goal of this thesis is to review and benchmark is therefore a systematic review and benchmark of existing methods for hybrid *de novo* assembly methods and their comparison with non-hybrid methods. The findings are then applied to create an improved transcriptome assembly for the cave salamander (*Proteus anguinus*).

1.2. Algorithmic basics of *de novo* transcriptome assembly

De novo nucleotide sequence assembly is a process that aims at the reconstruction of genomic or transcriptomic sequences from error-prone read fragments without reference. This requires

algorithms that account for the characteristics of biological sequences and are at the same time able to efficiently process the vast number of sequence reads needed to achieve sufficient coverage in sequencing experiments.

In this section, the two most common algorithmic approaches to the assembly of transcriptomic sequences are presented.

1.2.1 Overlap Layout Consensus assembly

Overlap Layout Consensus Assembly is an intuitive approach that includes three main stages, which also make up its name. In the first stage, the overlap stage, pair-wise read overlaps are computed. From the input reads an overlap graph is constructed according to definition 1. In the layout stage, stretches of paths with transitively-inferrible edges are identified. These stretches are then replaced by a consensus sequence which is computed by performing a multiple sequence alignments of the included sequences. Given the data at hand the consensus sequence is true nucleotide sequence. [15, 16]

After the third stage, an overlap graph constructed from transcriptomic reads would ideally consist of exactly one component per gene. In practice, these components can be fragmented due to insufficient read coverage or fused due to sequence homology. Graph components, representing alternatively spliced genes contain branching nodes. Transcript sequences are assembled by identifying sets of paths that collectively traverse all nodes in the overlap graph and explain the read coverage of the graphs nodes. This procedure is also showcased for a toy example in fig. 1.3.

Definition 1: Overlap graph

Given

1. a collection of reads $R = \{r_1, r_2, \dots, r_n\}$
2. a function $\mu : R^2 \rightarrow \mathbf{R}$ penalizing the best possible overlap for each read pair
3. a maximum value m

the corresponding overlap graph $OLG(R, \mu, t)$ is a directed and weighted Graph, where

1. Each read $r_i \in R$ is represented by a corresponding node n_i
2. Directed edges between pairs of nodes (n_i, n_j) exist, if $\mu(r_i, r_j) < m$
3. Edges are weighted by μ

1.2.2 De Bruijn graph assembly

As demonstrated in fig. 1.3, the de Bruijn graph (DBG) based approach starts by fragmenting the reads into fixed-length substrings (k -mers) from which a DBG according to definition 3 is constructed.

Definition 2: k -mer

Each k -mer is a string of fixed length k .

For a given sequence, its k -mers are the substrings of length k within that sequence.

Similarly, for a collection of sequences, its k -mers are the substrings of length k within those sequences.

Definition 3: De Bruijn graph

Given a collection of reads $R = \{r_1, r_2, \dots, r_n\}$, and a small integer number k , the vertices and edges of the corresponding de Bruijn graph $DBG(R, k) = (V, E, \mu)$ are determined by the k -mers of the reads.

$$V = \{(k-1)\text{-mer} \mid k\text{-mer is substring of one or more reads in } R\}$$

$$E = \{(v_i, v_j) \mid (v_i, v_j \in V) \wedge v_i \text{ is prefix and } v_j \text{ is suffix of a } k\text{-mer in } R\}$$

$$\mu : E \rightarrow \mathbf{N} \text{ (the weight } \mu(e) \text{ of an edge } e \text{ is the multiplicity of the corresponding } k\text{-mer in } R)$$

Note: This definition of the DBG follows its adaptation in bioinformatics and differs from the original formulation in graph theory.

Definition 4: Contig

A contig is an assembled continuous sequence.

It corresponds to a path in the overlap or de Bruijn graph.

Definition 5: Unitig

A unitig is an unambiguously assembled continuous sequence.

It corresponds to a non-branching path in the overlap or de Bruijn graph.

Addressing the double-strandedness of DNA (and cDNA, commonly used in RNA sequencing) the definition of the DBG is then extended so that that pairs of k -mers and their reverse complements form a single node [17, 18], usually only represented by the lexicographically smaller k -mer of the pair. [17]

Unlike for the Overlap Layout Consensus (OLC) approach, sequences are assembled by identifying paths that traverse all edges of the assembly graph instead of its nodes. [19]

As DBGs only store each k -mer once, they constitute a significant reduction of redundant information compared to the input read data. The graphs structure is affected by the read data in similar fashion as the overlap graph and additionally by the choice of the value of k . Small values for k may cause ambiguities in the assembly as repeat regions shorter than k are represented by a single edge which may cause additional branches and loops in the graph. Large values for k , on the other hand, may cause a fragmentation of the assembly graph at low coverage regions if there are regions where no pair of reads overlap at least in $k-1$ bases.

Based on topology and coverage information, DBGs are often edited to account for and remove sequencing artifacts, which also leads to a simplification of the graphs structure as demonstrated in fig. 1.2. [19, 20, 18, 21] These edit operations can be tailored to specific characteristics of the type of input sequencing data. [22] The resulting graph is often referred to as assembly graph. However, the terms de Bruijn graph and assembly graph are often used interchangeably. The memory footprint of de Bruijn and assembly graphs can be reduced by compression, i.e. collapsing nodes along non-branching paths into a single node, also shown in fig. 1.2.

From the de Bruijn graph, transcripts are assembled in similar fashion as from the overlap graph.

However, here the goal is to find sets of paths that traverse all edges of the graph.

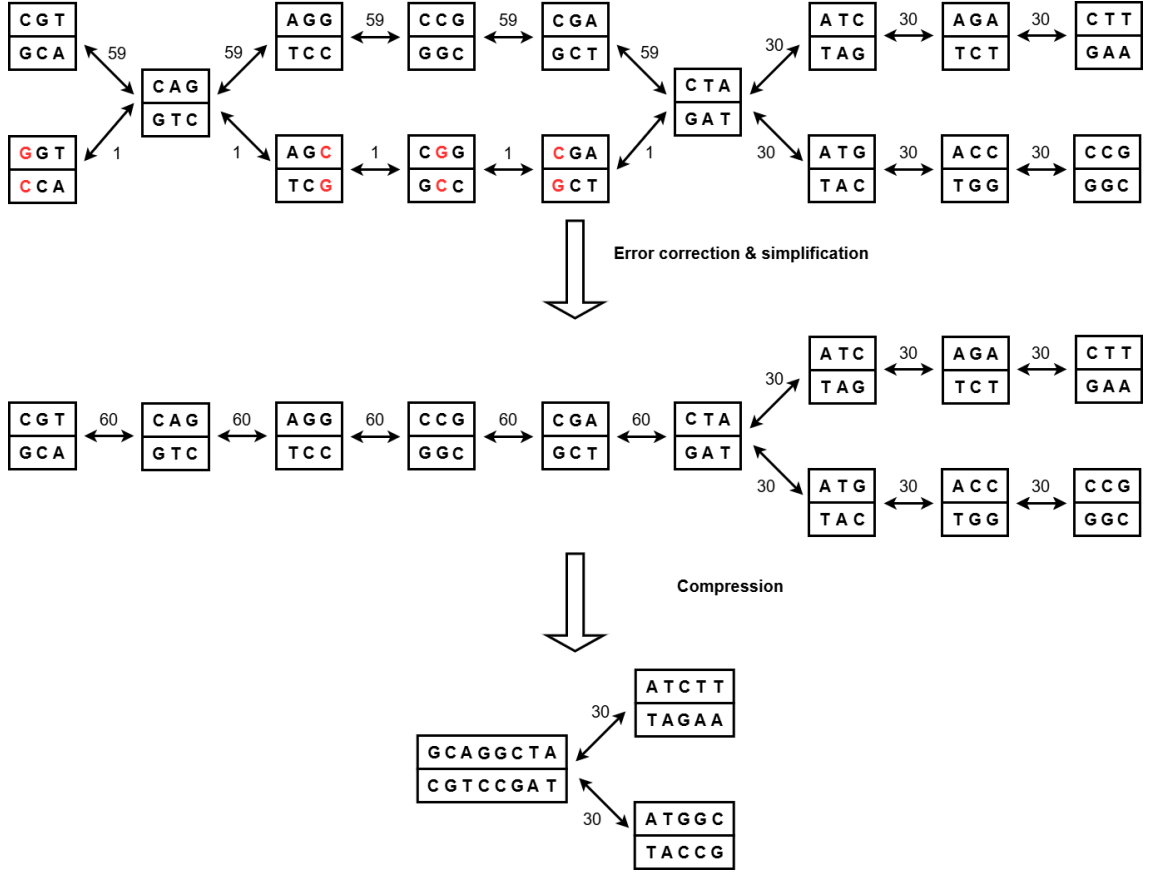


Figure 1.2: De Bruijn graph error correction and simplification:

In this toy example the cDNA of two short transcript isoforms (GCUGGCUA-GAA and GCUGGCUA-GGC), which use alternative exons at the 3'-end, were sequenced with an even read length of 4 and even coverage of 30. From them, a de Bruijn graph was constructed also using a k -mer length of 4. Its edges are annotated with the multiplicity of the corresponding k -mers in the read data. Two of the reads contained incorrect bases, highlighted in red, which result in a "bubble" (paths sharing the same start and end node) and an additional "tip" (A node with in-degree or out-degree 0). Even without knowledge of the true transcript sequences it is reasonable to assume that the additional paths stem from sequencing errors considering both are short and have a coverage of only 1. They are therefore removed.

Next, the non-branching paths are collapsed into single nodes.

Comparison of assembly approaches

Both the OLC approach and the DBG approach have their own advantages and disadvantages, which vary in importance depending on the input data.

The overlap layout consensus approach offers biological intuition and leverages the full read information. Building the overlap graph, however, requires pair-wise alignments and therefore time quadratic to the number of input reads which makes this approach impractical for the application with the large amount of Short Read (SR)s generated by modern massively parallel sequencing.

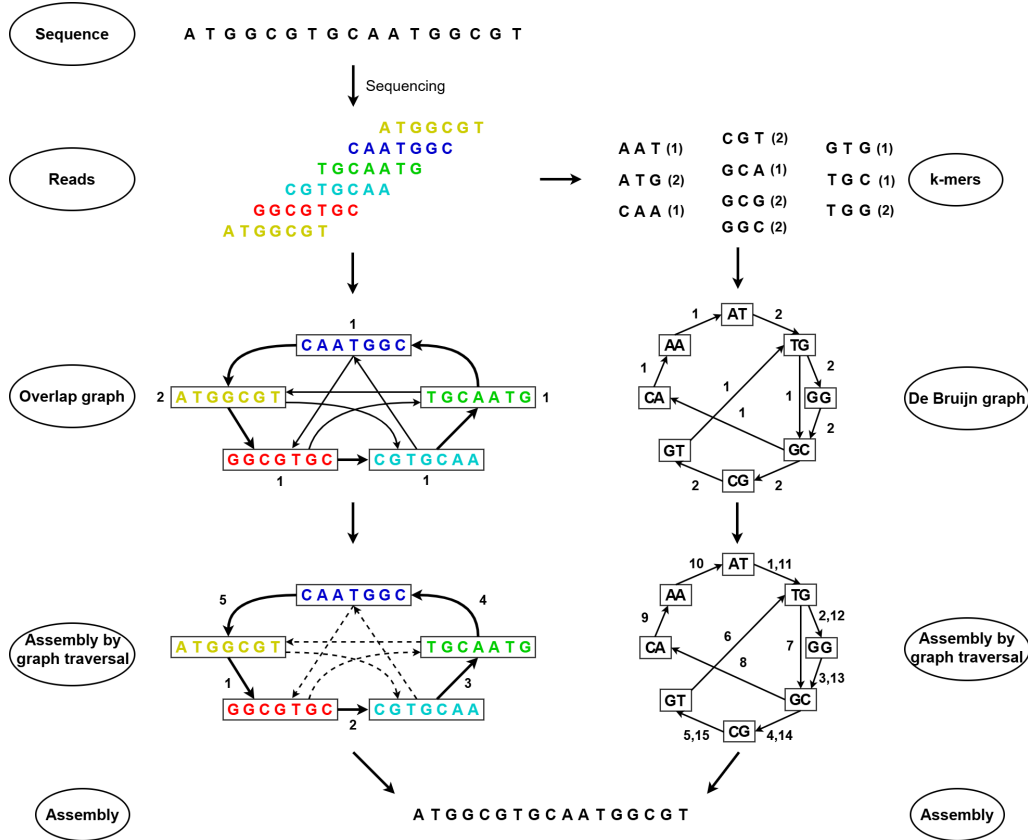


Figure 1.3: Toy example of *de novo* sequence assembly from error free sequence reads with even coverage.

Left: Overlap layout consensus approach.

In a first step the overlap graph is constructed. Arrows represent directed edges (u, v) that indicate perfect overlaps between reads where the suffix of read u is prefix of read v . Overlaps have lengths 1 (not shown), 3 (thin lines) and 5 (thick lines). Nodes are also annotated with their multiplicity. In a second step the optimal path that includes all nodes according to their multiplicity and maximizes overlaps is identified. The order of the edges in the path is indicated by the numbers next to them. The sequence is then assembled according to the identified path.

Right: De Bruijn-Graph approach.

First, *k*-mers of length 3 and their multiplicities are extracted from the sequence reads. Second, the de Bruijn graph is constructed from the *k*-mers. Here, edges correspond to the *k*-mers and their weight correspond to their multiplicity / coverage. Third, a path that traverses all edges according to their weight is identified. The order of the edges in the path is indicated by the numbers next to them. The sequence is then assembled according to the identified path.

OLC assembly has therefore been largely abandoned in favor of the more time efficient DBG assembly. [19, 23, 24]. DBGs can be constructed in linear time relative to the number of input reads. Nevertheless, the OLC approach was reintroduced for the assembly of transcriptomes from data obtained using the newer Single-Molecule-Real-Time (SMRT) sequencing methods which produces fewer but longer reads.[25]

1.3. Algorithmic strategies for hybrid transcriptome assembly

Based on the current literature, existing approaches that combine long and short reads in the context of *de novo* transcriptome assembly can be broadly divided into three categories. These approaches, shown in fig. 1.4 integrate the two read types at different stages of the assembly process. Firstly, long reads can be used to extend transcripts assembled from short reads, an approach implemented by IDP-denovo. [26] Secondly, by utilizing long reads to identify paths in a short read assembly graph that are likely to correspond to real transcripts, Trinity and SPAdes integrated the use of long reads into their pre-existing workflows. [27] Thirdly, long read assemblers such a RATTLE [28] or RNA-Bloom [29] can be combined with the use of more accurate short reads to remove errors in less accurate nucleotide sequences. This correction may happen pre assembly in the form of hybrid long read error correction or post assembly as transcript polishing.

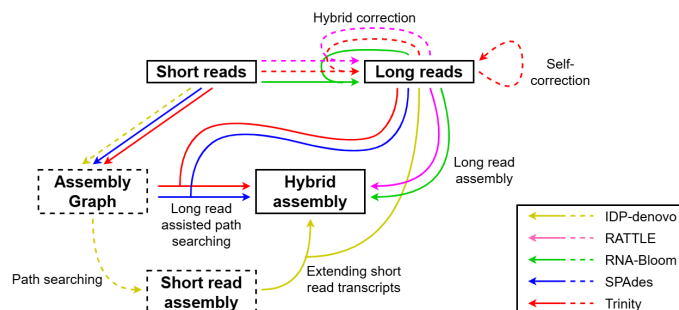


Figure 1.4: Road map for hybrid transcriptome assembly:

Steps for each tool are indicated by arrows in the respective colors. Arrows have solid lines if the tools are capable of performing the step on their own and dashed lines if additional software is required. Preceding error correction is neither demanded by IDP-denovo nor by SPAdes. The option to apply long read correction methods to transcripts assembled from LR has been omitted for reasons of clarity.

1.4. Review of existing tools for hybrid transcriptome assembly

1.4.1 Hybrid long read error correction tools

The two most common ways of categorizing long read correction methods are: a) according to the type of input data: In self-correction, long reads are utilized to correct each other while hybrid methods use additional accurate short reads to correct input long reads. Generally, better results can be expected from hybrid assembly methods. [14] b) according to their algorithmic approaches: Most of them fall into one of two categories. Firstly, in the multiple sequence alignment approaches, the reads used for the correction are recruited to the target long reads by first performing pairwise alignments followed by multiple sequence alignments for each target read and the recruited reads for correction. The corrected long read is given by the consensus sequence determined from the multiple sequence alignment. Secondly, in the DBG approaches, DBGs or assembly graphs are constructed from the reads used for the correction. Long reads are "anchored" to the graph by finding short exact or nearly exact matches (seeds) between the long reads and the sequences of the graph. Starting from those reads, the graph is traversed by extending the long read to graph alignment. The sequence corresponding to the path traversed in the graph with the best

alignment is used to correct the long read. Long read correction methods may output full length long reads (fl), trimmed long reads (t) where uncorrected ends are removed, or trimmed and split long reads (s) where uncorrected regions are removed both at the ends of and within long reads. Some long read correction methods perform trimming, which is the removal of uncorrected ends of the reads. Correction methods that perform read splitting also remove uncorrected regions within reads. DBG-based correction methods require less computational resources compared to multiple sequence alignment based methods. [30] Although most long-read error correction tools are intended for genomic reads, they generally are also applicable to transcriptomic reads. However, it has been reported that the sequences of the long reads are occasionally modified in such a way that they correspond to a different isoform instead of the original one. [14] Table 1.1 provides a non-exhaustive overview of software tools for hybrid longread correction.

1.4.2 Assembly Tools

Here, five software tools are described that can be utilized for hybrid transcriptome assembly, either on their own or with the help of additional tools.

- **IDP-denovo**

IDP-denovo attempts to improve precomputed transcriptome assemblies generated from short reads only by extending the transcript sequences by aligning long reads. The (extended) transcript sequences from the short read assembly are clustered based on information provided by the short read assembly method and the remaining unaligned long reads are clustered based on k -mers. Based on pseudo-reference sequences generated for each cluster an isoform-structure annotation is created and the transcript abundances estimated based on their short read and long read coverage. [26]

- **RATTLE**

A tool for the reference-free assembly and quantification of transcripts from nanopore long reads. RATTLE follows a four step process:

1. Gene level clustering:

RATTLE avoids an expensive all versus all mapping by using a greedy algorithm that sorts long reads into gene clusters. The clustering is an iterative process starting with each read forming a separate cluster. The n long reads of a cluster are stored in a list sorted in descending order by their length. For each cluster the read with rank $n \times 0.15$ (This value can be adapted by the user) serves as representative element of the cluster. Pairs of clusters whose representative elements have a RATTLE similarity above a defined threshold are merged in a greedy manner. This threshold is decreased after each step down to a certain level. For pairs of long reads that represent read clusters and share sufficiently many k -mers of length 6 the RATTLE similarity is computed as the fraction of bases of co-linear matching k -mers in the shorter read.

2. Transcript level sub-clustering:

Based on the gap-length distribution of the previously calculated co-linear matching k -mers the reads of the gene clusters are further divided into transcript isoform clusters.

3. Long read error correction:

Consensus sequences determined from multiple sequence alignments for every transcript isoform cluster are used for the correction of long reads.

4. Transcript polishing and quantification:

Using the algorithm from step 1, transcript clusters might be merged again. A final multiple-sequence alignment determines the transcript sequence, and the transcript abundance is given by the number of reads in the cluster.

Tool	Type	Output			Algorithmic idea
		fl	t	s	
FMLRC	DBG	✓			Leverages FM-indices from the Burrows-Wheeler Transform to represent DBGs for arbitrary k -mer lengths. Dynamically adjusts k -mer lengths per read: smaller for precise anchoring, larger for less complex graph traversal. [31]
FMLRC2	DBG	✓			Near identical Rust re-implementation of FMLRC. [32]
HALC	DBG	✓	✓	✓	Long Read (LR) correction by traversal of a contig graph where nodes represent regions of contigs generated from SRs and edges and their weight reflect the frequency of neighboring LR regions mapping to pairs of contig regions. [33]
Jabba	DBG			✓	The corrected sequence is dictated by a path in an assembly graph constructed from SRs with maximal exact matches between LR and the nodes along this path. [20]
LoRDEC	DBG	✓	✓	✓	Constructs a DBG only from "strong" k -mers with sufficient coverage compared to "weak" k -mers with insufficient coverage. Regions in the LRs are replaced with the sequences of the paths of the DBG they were mapped to. [30]
Proovread	MSA	✓	✓		LRs are corrected by iteratively performing multiple sequence alignments for individual LRs and aligning SRs within windows of increasing size. [34]
Racon	MSA	✓			Consensus sequences determined by multiple sequence alignments using partial order alignment graphs constructed from pairwise read mappings. [35]
Ratatosk	DBG	✓		✓	Correction of LRs for variant calling using a DBG colored with the SRs contributing the respective k -mers. A confidence threshold prevents false corrections of true biological variations [17]
RNA-Bloom	DBG		✓		LR <i>de novo</i> transcriptome assembler with an integrated LR correction module that uses a DBG constructed from k -mers of those LRs and optionally additional SRs. Similar to LoRDEC, RNA-Bloom only uses strong k -mers. However, addressing the varying transcript expression levels, RNA-Bloom dynamically computes local thresholds to distinguish weak and strong k -mers during the extension of the read to graph alignments. [25]
TALC	DBG		✓		Transcript-level Aware LR Correction. After LR anchoring a breadth-first search identifies its corresponding path in the DBG by minimizing the edit distance between path and read. At branching nodes only sets of nodes whose coverage is consistent with that of the current path are considered. TALC explicitly takes the typically lower complexity of untranslated regions into account. [36]

Table 1.1: Summaries for a selection of hybrid long read error correction methods

[28]

Although RATTLE is intended for long read only assemblies it can be used for hybrid assembly by combining it with hybrid long read error correction methods.

- **RNA-Bloom**

RNA-Bloom is a software tool that includes transcriptome assembly from long reads. short reads can additionally be provided for hybrid error correction. It follows a six step process and uses Minimap2 [37] to compute alignments at various steps:

1. Long read correction:

RNA-Bloom can correct long reads using simplified Bloom filter de Bruijn graphs constructed from the provided read data. Bloom filter are space-efficient, probabilistic data structures for storing sets of messages (here, k -mers) and enable fast membership checking for potentially not yet included messages [38] which allows rapid de Bruijn graph construction.

2. Digital normalization:

In order to reduce the computational resources required in the later assembly steps, a minimal longest reads set is sampled from the long reads. Long reads are first sorted by their length and iteratively added to the minimal longest reads set if their coverage estimated by counting strobemers does not yet exceed the set target depth. Strobemers are lists of very short k -mers (strokes) occurring successively but with variable spacing, making them suitable for similarity detection of long reads with higher rates of insertions and deletions. [39]

3. Long read trimming / splitting: Regions with insufficient coverage are trimmed from the long reads. Long reads are also split at regions with low coverage as these could be chimeric reads stemming from accidentally fused transcripts.

4. Unitig assembly:

RNA-Bloom constructs an overlap graph from the long reads and assembles unitigs.

5. Unitig polishing:

The corrected long reads are mapped to the unitigs. Using the resulting mappings, remaining errors in the unitig sequences are removed by Racon [35].

6. Transcript assembly:

Transcripts are assembled from an overlap graph of the polished unitigs.

[25]

- **SPAdes**

SPAdes is a de Bruijn graph-based assembly tool originally designed for the assembly of genomes from short reads from single-cell sequencing only. In order to both disentangle the assembly graph, specifically at repeat regions, while also avoiding fragmentation at regions with lower coverage, SPAdes uses multisized de Bruijn graphs, i.e. the union of de Bruijn graphs for different k -mer lengths. [21] Later, it has been extended to perform hybrid assemblies. By mapping long reads to the short read assembly graph, SPAdes attempts to better identify paths in the assembly graph that correspond to true biological sequences and to close coverage gaps in the assembly graph by using consensus sequences generated from the long reads. [27]. For transcriptome assembly, the assembly graph simplification and error removal algorithm used by SPAdes has been adapted to the specific characteristics of RNAseq data. [22]

- **Trinity**

Similarly to SPAdes, Trinity is a de Bruijn graph-based assembly tool. However, it has been designed for transcriptomes from the beginning and takes a unique approach to the construction of the de Bruijn graph. Its three-step procedure is realized by separate sub-modules.

1. Greedy contig assembly:

Inchworm creates a dictionary with k -mers and their frequencies. After removing k -mers that likely contain errors, transcript contigs are generated. The contig assembly starts with the k -mer that has the highest frequency. It is then bidirectionally extended. Forward it is extended using a k -mer whose first $k-1$ bases are identical to the last $k-1$ bases of the contig, always choosing the k -mer with the highest frequency. Similarly, the contig is extended in the other direction using a k -mer whose last $k-1$ bases are identical to the first $k-1$ bases of the contig, also always choosing the k -mer with the highest frequency. k -mers incorporated into a contig are removed from the dictionary. This process is repeated until the dictionary is empty.

2. De Bruijn graph construction

Chrysalis constructs the de Bruijn graph components from the Inchworm contigs using their $k-1$ bases long exact overlaps.

3. Butterfly first simplifies the graph structure to remove errors. It then scores alternative paths in the assembly graph according to their read / read pair support and assembles transcripts accordingly.

[40]

Trinity has also been extended to be able to use long reads to identify paths in the assembly graph.

1.5. Transcriptome assembly quality control

The *de novo* assembly of an organisms transcriptome is a multi-step process that includes the collection of the biological sample, the preparation of an RNA library, its sequencing, the pre-processing of the read data and the computational reconstruction of transcripts from the RNA sequence reads. Each of these steps involves potential sources of errors and technical limitations that may influence the final transcriptome assembly. Data from RNA sequencing experiments contain biological artifacts: transcripts that are already fragmented in the cells, contain residual introns from incomplete splicing or belong to microbial organisms that colonise the organism under investigation. Data from RNA sequencing experiments also contain technical artifacts: Fragmentation of transcripts during library preparation, chimeric sequence reads from transcripts, or transcript fragments, that were unintentionally ligated, sequence reads from foreign transcripts from sample contamination and base-calling errors. These artifacts cannot be completely eliminated in the assembly process. Furthermore, transcripts are often only captured partially or not captured at all by the sequencing data, which leads to missing or incomplete transcripts in the final assembly. Furthermore, the fusion or fragmentation of transcript sequences may also be caused by the assembly process itself, as described in section 1.2.2. Consequently, robust methods for the evaluation of the quality of transcriptome assemblies are required.

1.5.1 Reference-free methods

Evaluating *de novo* assemblies of poorly studied organisms is particularly difficult because the biological ground truth is often limited or entirely unknown.

Measuring transcript continuity

An older but still commonly used measure for evaluating the continuity of assembled nucleotide sequences is the N50 value. Given a set of sequences, the N50 value is the maximum length of a sequence such that all sequences in the given set with lengths greater or equal to N50 contain 50% of the total nucleotides. [41] Similarly, Nx values can be defined for any other percentage.

Genomes or transcriptomes tend to be better covered by assemblies with higher Nx values. However, it must be taken into account that the Nx value does not yet say anything about the correctness of the individual sequences or the completeness of the assembly. By not reporting shorter sequences or incorrectly fusing them into larger ones, assembly tools can increase Nx values despite creating less accurate assemblies.

BUSCO completeness

Within larger phylogenetic clades, some highly conserved genes can be identified that are present in the majority of the genomes of species belonging to that group. Scanning genome or transcriptome assemblies for the presence of these Single-Copy Benchmarking Orthologs provides an evolutionary informed estimation of their completeness without the need for a reference. [42]

TransRate

TransRate checks the coherence of assembled transcript sequences with their input data. For each transcript, TransRate computes a variety of score components. These evaluate the coverage of the transcripts and the read support of their bases, as well as their structure. Based on these score components, a contig score is calculated for each assembled transcript. These contig scores allow to assign an overall assembly score as well as the filtering of implausible transcripts. [43] However, it is no longer maintained and does not support the input of long reads.

1.5.2 Reference-based methods

Assembled transcriptomes can also be examined by comparing them with an annotated reference genome, if one is available. This also allows to evaluate *de novo* methods using data from model organisms with near complete and accurate genome annotations before applying them to new organisms. Transcripts in *de novo* assemblies generated from reads that are either simulated directly from the reference or stem from an individual of the same inbred line should ideally show no or little deviation from the reference.

rnaQUAST

In addition to some more basic measures such as the N50 value and length distribution of the input transcripts, rnaQUAST provides detailed metrics on the quality and completeness of *de novo* transcriptome assemblies from transcript to genome alignments.

It is reported how many and what proportion of assembled transcripts can be aligned with the genome and are matched by exons (see fig. 1.5). Transcripts that do not match any isoform are reported as unannotated. Transcripts that, according to their best scored spliced alignments with the genome, have parts that a) map to very distant genomic loci on the same or different chromosomes, b) do not stem from the same chromosomal strand or c) contradict each other in their orientation, are reported as misassemblies. For genes and their isoforms, the proportion covered by a single assembled transcript (% assembled) or by multiple transcripts (% Covered) is also specified. [44]

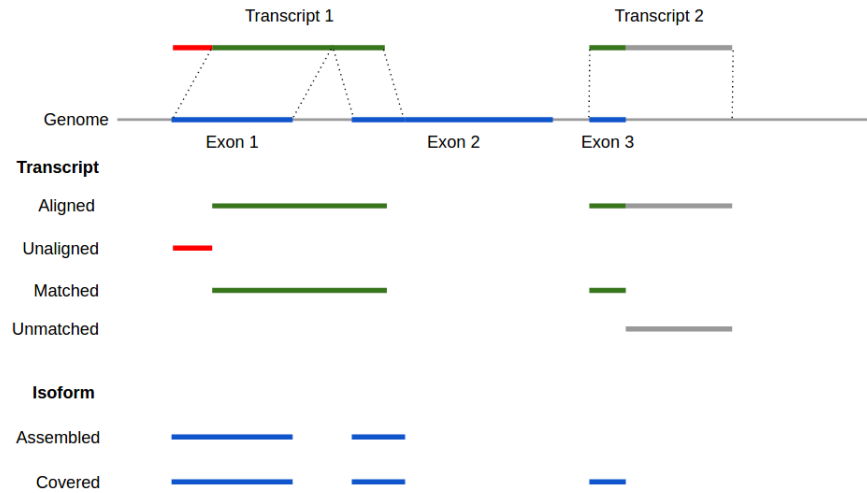


Figure 1.5: This figure from the rnaQUAST repository illustrates the quality metrics generated by rnaQUAST. helps in understanding the transcripts and isoform metrics in rnaQUAST outputs.

The shown stretch of the genome contains a gene with an isoform consisting of 3 exons with two assembled transcripts aligned to it.

Transcript metrics: rnaQUAST determines whether and to what extent assembled transcripts could be aligned with the genome and match exons.

Isoform metrics: rnaQUAST determines whether and to what extent reference isoforms are covered, i.e. having one or multiple corresponding assembled transcripts, or are assembled, i.e. having a single longest corresponding assembled transcript.

SQANTI3

SQANTI3 characterizes the structural isoforms of long read defined transcriptomes in relation to a provided genome annotation. It can be used for quality control, filtering of potential artifacts and discovery of novel isoforms. [45]

2. Methods and Materials

All software tool used in this thesis and their versions are listed the appendix in table 1

2.1. Test data

Long read error correction and assembly methods were tested with publicly available Illumina short reads and matching simulated long reads of the model organism *mus musculus*. Throughout this thesis, the annotated genome assembly GRCm39 (strain C57BL/6J) was used as reference

2.1.1 Real Illumina short reads

A total of 58,219,222 Illumina sequence read pairs from six tissues (brain, heart, kidney, liver, lung, and stomach) of the mouse (strain C57BL/6JJcl) were obtained from the "Comprehensive Mouse Transcriptomic BodyMap across 17 Tissues by RNA-seq" [46]. The downloaded samples with SRA accession numbers SRR5273687, SRR5273695, SRR5273699, SRR5273700, SRR5273704 and SRR5273705 are stored in the NCBI-SRA as part of the bioproject PRJNA375882. All reads were pooled into a single file.

Only for the testing of the long read error correction chromosome 1 read pairs were sampled. For this purpose, the IDs of genomic features whose type was 'transcript' or contained 'RNA' were first extracted from the reference annotation, followed by the corresponding transcript sequences from the reference transcriptome. The read pairs were then filtered by aligning them to the chromosome 1 transcripts with Minimap2.

2.1.2 Simulated nanopore long reads

From the pooled short reads, transcript abundances were estimated with Kallisto [47] and its output converted with kallisto2nanosim from SSfSBT. Using the estimated transcript abundances and a statistical model of a sequencing process learned from PromethION nanopore long reads, publicly available at NCBI-SRA under SRA accession number SRR11517552, reads were simulated with NanoSim [48] as shown in fig. 2.1.

For the evaluation of the assembly tools, sets of five and ten million long reads were generated. For the testing of the error correction methods, transcript abundances of non-chromosome 1 transcripts were explicitly set to 0 in the respective file and one million chromosome 1 long reads were simulated.

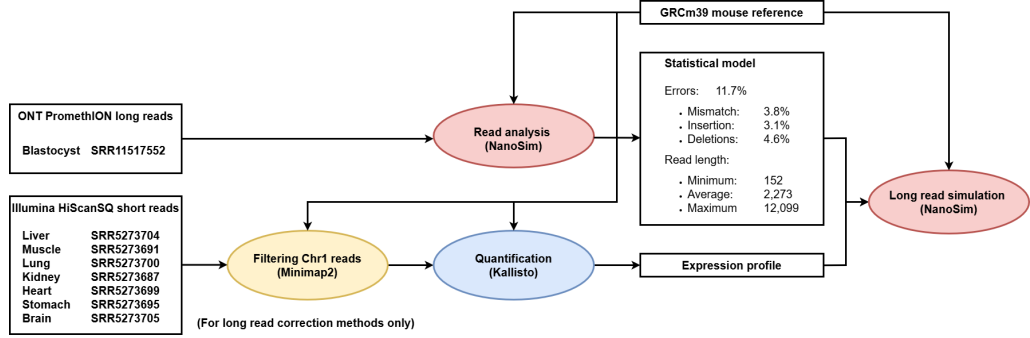


Figure 2.1: Creating sets of matching mouse short and long reads: Short reads from six mouse tissues were downloaded from the NCBI. For the specific evaluation of long-read correction tools, a subset of short reads was sampled by aligning them to chromosome 1 reference transcripts. Transcript abundances were estimated using Kallisto. A statistical model was trained using mouse blastocyst PromethION reads (also obtained from the NCBI) with NanoSim, which was then used to simulate long reads based on the estimated transcript abundances.

2.2. Benchmark 1: Hybrid Long Read Correction

2.2.1 Software tools and parameter settings

Eight tools for hybrid error correction were considered in this benchmark: FMLRC2, HALC, Jabba, LoRDEC, Proovread, Ratatosk, RNA-Bloom, and TALC.

FMLRC2 requires a precomputed FM-index of the short read data which was constructed with Ropebwt2 [49]. HALC was run in ordinary mode and was provided with the sequences from an assembly graph generated with SPAdes. For Jabba, a de Bruijn graph with a k -mer length of 31 was built using Brownie, which is bundled with the Jabba distribution. Jabba outputs corrected and uncorrected separately. Uncorrected reads were excluded from the evaluation. LoRDEC was run with parameters ‘-k 21’ and ‘-c 3’ (DBG construction from k -mers of length 21 that appear at least three times in the short reads). Ratatosk was run with parameter ‘-m 0’ (sets the minimum confidence threshold to correct a SNP to 0). RNA-Bloom was run with parameter ‘-stage 2’ (terminates after correction and before assembly) both in hybrid and in non-hybrid mode.

2.2.2 Evaluation

Alignment analysis

Raw and error corrected long reads were aligned with the reference genome by Minimap2. The alignments were converted to BAM format and sorted, and indexed using Samtools [50]. The alignments were then evaluated with AlignQC [51]. Additionally, error rates were also computed with Cramino [52] and NanoSim.

Expression profiling

NanoSim was used to estimate transcript abundances from read-to-transcriptome alignments. The analysis of the expression profiles was performed with a custom jupyter notebook which can be found in the supplementary data.

2.3. Benchmark 2: Assemblies

2.3.1 software tools and parameter settings

Non-hybrid *de novo* assemblies for comparison

De novo short read assemblies were generated with Velvet+Oases [18, 53], RNA-Bloom, Trinity, and SPAdes. For Velvet+Oases, k -mer lengths of 23 and 31 were selected. SPAdes was run with the ‘-rna’ parameter. *De novo* long read assemblies were constructed with RATTLE and RNA-Bloom using default parameters.

Hybrid *de novo* assemblies

The two sets of five and ten million simulated nanopore long reads were hybrid corrected by RNA-Bloom, LoRDEC with trimming and FMLRC2 in combination with LoRDEC, utilizing the Illumina sequence read pairs downloaded from NCBI-SRA. The corrected long reads were then used to construct hybrid *de novo* assemblies. For each of the six sets of corrected long read assemblies were generated with IDP-denovo, RATTLE, RNA-Bloom, SPAdes and Trinity. IDP-denovo was additionally provided with the previously created Velvet+Oases assembly, SPAdes and Trinity with the short reads. SPAdes was again run with the ‘-rna’ parameter and RNA-Bloom with ‘-e 0’, setting the number of iterations of error correction to 0. All other tools were run with default parameters. Furthermore, LoRDEC was applied to the transcripts assembled by RNA-Bloom from 5 million uncorrected long reads for transcript polishing.

Reference-based StringTie assemblies for comparison

For each dataset from which a *de novo* assembly was created, a corresponding reference genome was also created. For this purpose, the reads were aligned with the genome by STAR [54] for the short read pairs and Minimap2 for the long reads, respectively. Splice-site hints from the reference annotation were used neither for the mapping of the reads nor for the assembly with StringTie [55, 56].

2.3.2 Evaluation

Extraction of StringTie transcript sequences

Transcript sequences of the StringTie assemblies, which are output in GTF-format were extracted from the reference genome assembly with GffRead [57]

BUSCO Completeness

The completeness of vertebrate BUSCOs within each assembly was assessed with BUSCO using the OthoDB v10. [58]

rnaQUAST quality control

The quality of each assembly also was evaluated from GMAP [59] transcript-to-genome alignments with rnaQUAST. The source code of rnaQUAST was adapted to capture all transcript types of the reference annotation. The changes are described in detail in the appendix (section .3). The results from this analysis were plotted with rnaQUASTcompare from SSfSBT.

2.4. Assembly of the *Proteus anguinus* transcriptome

2.4.1 Sequencing and preprocessing

Paired-end Illumina short reads and MinION nanopore long reads of one individual (p4) were sequenced as described in the publication on our preliminary olm transcriptome assembly. [10] Raw reads can be obtained from bioproject PRJNA1207464. The preprocessing of the Illumina short reads is also described in the pre-print. Additionally, adapter sequences were removed from the Nanopore long reads using Porechop [60].

2.4.2 Long read self-correction and assembly

Long read self-correction and assembly was performed without short reads with RNA-Bloom [25].

2.4.3 Contamination filtering and hybrid transcript polishing

Potential human and microbial were removed from the Illumina short reads with Kraken2 [61] using the standard database. The assembled transcripts were then polished with LoRDEC [30] with parameters ‘-k 23’, ‘-s 4’ (DBG construction from *k-mers* of length 23 that appear at least four times in the short reads). With `lr_lordec_contam_filter` transcripts that were not aligned to the short read de Bruijn graph by LoRDEC were removed.

2.4.4 Supplementing protein-coding transcripts from previous assembly

The preliminary transcriptome assembled with Trinity and the new transcriptome assembled with RNA-Bloom were mapped to the proteins of the reviewed UniProt Knowledgebase [62] with DIAMOND [63]. For proteins, where DIAMOND did not report a hit in the new RNA-Bloom assembly, but in the old Trinity assembly, the Trinity transcript with the best hit (lowest e-value) was selected and added to the new RNA-Bloom assembly.

In a similar manner, transcripts were added for vertebrate BUSCOs that were still missing. The BUSCO tool [42] was run on both assemblies using the OthoDB v10 [58]. For BUSCOs, where the BUSCO tool did not report a hit in the RNA-Bloom assembly, but in the Trinity assembly, the Trinity transcript with the best hit (lowest e-value) was selected and also added to the new RNA-Bloom assembly.

2.4.5 Removal of potential frame-shift errors

Proofframe [64] was employed to detect and correct potential insertion and deletion errors within coding sequences that would cause shifts of the open reading frames.

2.4.6 Functional annotation

Open reading frames were extracted using TransDecoder [65]. The eggNOG-mapper [66] with additional parameters ‘-m diamond’, ‘-itype CDS’, ‘-sensmode sensitive’, ‘-dmnd_iterate yes’ and ‘-tax_scope auto’ and the eggNOG database v5.02 were then employed for their functional annotation.

2.4.7 p53 characterization

The p53 sequence of the olm was compared with that of other amphibians using multiple sequence alignment calculated and visualised with MAFFT online [67] using standard parameters.

The aminoacid sequence from another *Proteus anguinus* individual based on RNA sequencing data

2.4. ASSEMBLY OF THE PROTEUS ANGUINUS TRANSCRIPTOME

from Irisarri et al. [9], which was assembled with Trinity, was obtained from the supplementary data of Barta et al., 2017 [68]. From NCBI, sequences AEI91929.1 (*Hypselotriton orientalis* / chinese fire belly newt) and ABI33869.1 (*Ambystoma mexicanum* / axolotl) were downloaded and sequences A0A6P8Q2I6 (*Geotrypetes seraphini* / gaboon caecilian), A0A6P7WJT7 (*Microcaecilia unicolor*), A0A8J1MU12 (*Xenopus laevis* / african clawed frog) and Q6NTF1 (textitXenopus tropicalis / western clawed frog) from UniProt.

The protein structure of p53 for the olm and the axolotl was predicted and visualized with AlphaFold3 [69].

3. Results

Software tools for hybrid long read correction and transcriptome assembly were tested with RNA sequencing data from six tissues of the mouse. Illumina short reads were downloaded from NCBI-SRA, matching nanopore long reads were simulated with NanoSim. For the first benchmark of the long read correction methods, chromosome 1 transcript short reads were filtered in order to allow the testing of a large variety of tools within a reasonable time frame. Here, one million nanopore long reads were simulated. For the second benchmark of the transcriptome assembly tools, the full set of Illumina short reads was used and sets of five and ten million long reads were simulated. Both hybrid and non-hybrid transcriptome assemblies were created for all assembly tools. Three of the correction methods that stood out in the previous benchmark were used to correct the long reads for the hybrid assemblies.

3.1. Validation of the Benchmark Dataset

3.1.1 Short reads were filtered while preserving chromosome 1 transcript counts

From the 58,219,222 input Illumina short read pairs 3,678,636 ($\sim 6.32\%$) were sampled by mapping them to 7752 chromosome 1 transcript sequences from 2292 genes, with 1141 of them having alternative splice-isoforms. Figure 3.1 demonstrates that a) transcript counts for chromosome 1 transcripts were largely preserved and b) transcript counts for non-chromosome 1 transcripts were substantially reduced.

3.1.2 Simulated nanopore long reads with matching expression profile

The expression profile of the long reads simulated by NanoSim closely matches that of the filtered Illumina short reads (see fig. 2.1 c))

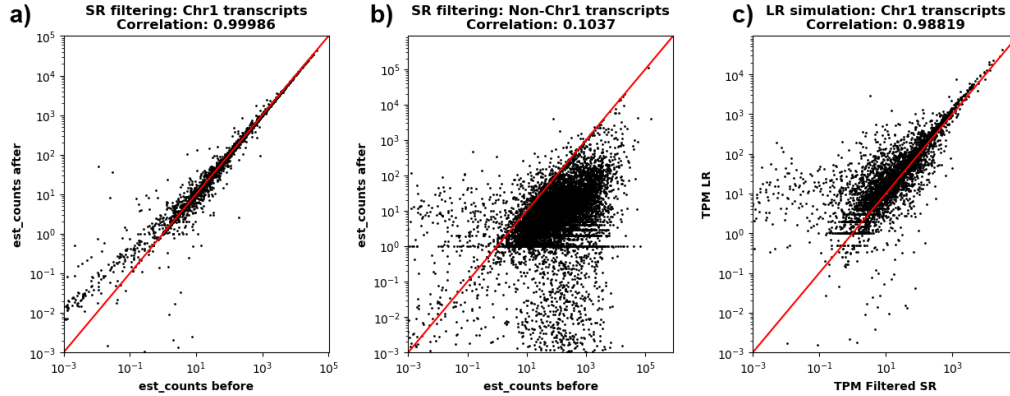


Figure 3.1: Evaluation of the dataset creation:

- Short read counts from Kallisto for chromosome 1 transcripts before and after reads were filtered by mapping them to chromosome 1 transcripts with Minimap2.
- Short read counts from Kallisto for non-chromosome 1 transcripts before and after reads were filtered by mapping them to chromosome 1 transcripts with Minimap2.
- Normalized expression values of chromosome 1 transcripts estimated with Kallisto from short reads and estimated with NanoSim from long reads that were also simulated with NanoSim.

3.2. Benchmark 1: Hybrid Long Read Correction

With approximately 3.7 million genuine Illumina short read pairs and one million matching simulated Nanopore long reads of chromosome 1, eight software tools for hybrid long read correction were tested individually and some also in combination. While some tools output corrected reads in full length, others also or exclusively output trimmed or split reads, in which uncorrected ends or even middle regions are removed. One tool (RNA-Bloom) was tested both with hybrid (h) and non-hybrid (n) correction. In total, 15 sets of corrected long read data sets were created and compared with each other.

TALC could not be compiled by following the instructions of the authors and was therefore excluded from the analysis. The authors of TALC do not provide a precompiled version of their tool and abandoned their project after its publication.

3.2.1 Alignment analysis

Analysis of read to genome alignments was performed with AlignQC. The error rate calculated by AlignQC for the uncorrected reads differed significantly from that reported by NanoSim for the long reads used to train the simulation model. The error rates in each read set was therefore also computed with NanoSim and Cramino.

Correction trims, splits or removes reads entirely

The results in of the alignment analysis presented in table 3.1 include basic information of the number of reads and bases in each evaluated long read dataset. Tools that output full length corrected long reads tend to return all input reads. Only Proovread returned 427 less reads. The output long reads often contain more bases than the input bases. Trimming lead to the removal of 9 (RNA-Bloom in non-hybrid mode) to 2521 (LoRDEC) long reads and reduced the number of bases by 6.5% (HALC) to 12.5% (LoRDEC). RNA-Bloom also uses k-mers from long reads during hybrid correction and is therefore able to correct long read regions without short read coverage.

CHAPTER 3. RESULTS

Tool	Reads	Aligned reads		Bases		Aligned bases		% of total reads	% of aligned reads
	total	total	%	total	aligned reads	total			
Raw reads									
-	1,000,000	990,705	99.1	1,443,239,648	1,440,179,969	1,219,533,257	84.5	84.7	
Full length reads									
FMLRC2	1,000,000	996,495	99.6	1,447,434,970	1,446,177,141	1,310,827,886	90.6	90.6	
+ LoRDEC	1,000,000	997,879	99.8	1,449,031,032	1,448,202,598	1,317,578,427	90.9	91.0	
HALC	1,000,000	999,055	99.9	1,464,965,219	1,464,623,851	1,288,398,509	87.9	88.0	
LoRDEC	1,000,000	997,836	99.8	1,459,675,670	1,458,893,512	1,274,271,414	87.3	87.3	
Proovread	999,573	991,030	99.1	1,445,257,661	1,442,595,470	1,237,469,417	85.6	85.8	
Ratatosk	1,000,000	992,334	99.2	1,439,072,372	1,436,717,899	1,292,439,998	89.8	90.0	
Trimmed reads									
HALC	999,867	998,354	99.8	1,348,884,260	1,348,647,814	1,278,968,413	94.8	94.8	
LoRDEC	997,479	994,498	99.7	1,263,040,669	1,262,856,999	1,259,739,341	99.7	99.8	
RNA-Bloom (n)	999,991	993,758	99.4	1,314,141,804	1,312,241,270	1,239,684,661	94.8	94.5	
RNA-Bloom (h)	999,988	993,103	99.3	1,309,225,899	1,307,114,499	1,239,358,550	94.7	94.8	
+ LoRDEC	999,988	997,594	99.8	1,312,505,503	1,310,673,842	1,248,313,362	95.1	95.2	
Trimmed and split reads									
HALC	1,154,199	1,150,369	99.7	1,275,200,520	1,274,735,265	1,262,189,512	99.0	99.0	
Jabba	826,971	684,480	82.8	532,780,281	524,400,067	523,938,632	98.3	99.0	
LoRDEC	1,146,361	1,143,656	99.8	1,246,825,769	1,246,462,068	1,244,156,554	99.8	99.8	
Ratatosk	2,400,735	1,997,797	83.2	1,027,846,312	992,499,767	982,270,988	95.5	99.0	

Table 3.1: Statistics reported by AlignQC on alignments of long read with the reference genome by Minimap2.

Long read sets are grouped into: raw reads (uncorrected), full length reads (corrected), trimmed reads (corrected, with uncorrected ends removed) and trimmed and split reads (corrected, split into their corrected parts). RNA-Bloom was tested in non-hybrid (n) and hybrid mode (h) Results of alignment analysis from AlignQC

Tool	% Errors		
	AlignQC	NanoSim	Cramino
Raw reads			
-	10.51	11.29	8.09
Full length reads			
FMLRC2	0.98	2.97	1.23
+ LoRDEC	0.42	2.15	0.41
HALC	0.39	1.98	0.45
LoRDEC	0.90	1.96	1.18
Proovread	7.86	8.78	6.58
Ratatosk	3.11	5.70	3.53
Trimmed reads			
HALC	0.39	1.86	0.38
LoRDEC	0.74	1.82	1.05
RNA-Bloom (n)	1.85	3.90	1.75
RNA-Bloom (h)	1.56	3.22	1.57
+ LoRDEC	0.36	2.25	0.51
Trimmed and split reads			
HALC	0.19	1.39	0.24
Jabba	0.01	1.15	0.00
LoRDEC	0.50	1.67	0.89
Ratatosk	0.55	5.38	0.79

Table 3.2: Base error rate computed from alignments by AlignQC, Cramino and NanoSim from alignments with the reference genome.

Long read sets are grouped into: raw reads (uncorrected), full length reads (corrected), trimmed reads (corrected, with uncorrected ends removed) and trimmed and split reads (corrected, split into their corrected parts). RNA-Bloom was tested in non-hybrid (n) and hybrid mode (h)

The error rates for NanoSim, broken down by substitution, insertion and deletion, can be found in fig. 3.2 and supplementary table 1.

Splitting further increased the number of removed bases. HALC and LoRDEC split only a minority of reads and removed only 11.6% and 13.6% of the input bases. Jabba failed to correct 409,623 of the input long reads. The remaining 590,377 reads were split into 826,971 fragments. In total, Jabba removed 63% of the bases. The split reads from Ratatosk were highly fragmented and had 28.8% of the input bases removed.

Correction improves mappability

All correction tools that output full length or trimmed long reads improved the mappability compared to the uncorrected reads. Both the number of aligned reads and the number of aligned bases were increased by the correction (see table 3.1). The highest percentage of mapped reads was achieved with the HALC corrected long reads. FMLRC2 is the tool for which the highest number of bases could be aligned. By additionally correcting them with LoRDEC, this number was further increased. Similarly, the mappability of the RNA-Bloom corrected reads was also improved by additionally correcting them with LoRDEC. Long read trimming and splitting generally increased the fraction of mapped reads and bases at the cost of removed bases. The only two exceptions here are the highly fragmented split reads from Jabba and Ratatosk.

AlignQC, Cramino and NanoSim report different base error rates

AlignQC, Cramino and NanoSim report different error rates (see table 3.2). Some aspects of the respective error rate calculations are only vaguely described in the corresponding publications, but can be mostly understood by examining the source code. All tools consider only the best alignments for a given read. While both Cramino and NanoSim take all such alignments into account, AlignQC reads alignments until a total alignment length of at least one million bases is reached. This sampling leads to less accurate error rates that can change on the number of threads used during the alignment step (see table 2). Furthermore, AlignQC, Cramino and NanoSim score insertions and deletions differently. Cramino for example computes a gap-compressed identity, treating the insertion or deletion of multiple nucleotide bases as a single error [52]. NanoSim on the other hand counts each inserted or deleted base separately. The similarities of the error rates reported by AlignQC and Cramino suggest they use the same scoring method.

Correction reduces base error rates

The base error rates are listed in table 3.2 and for NanoSim and Cramino also shown in fig. 3.2. The NanoSim error rate for the raw reads is 11.29%. The error rates of the corrected full length reads range from 1.96% (LoRDEC) to 8.78% (Proovread). The error rates for trimmed and split reads range from 1.86% (LoRDEC) to 3.22% (RNA-Bloom) and 1.15% (Jabba) to 5.38% (Ratatosk). As expected, the removal of uncorrected regions by read trimming and splitting reduces the error rates. The error rates calculated by AlignQC and Cramino are significantly lower than those calculated by NanoSim. As shown in fig. 3.2, sorting the tools according to their error rate would result in a different order for Cramino and NanoSim. In particular, LoRDEC, for which the corrected reads contain more mismatch errors than insertions and deletions, is ranked lower according to the Cramino error rates.

3.2.2 Expression profiling

An earlier study by Lima et al. showed that long read corrections can alter the expression profile and that individual long reads can take on different isoforms. [14] This may have a negative impact on the quality of transcriptomes assembled from corrected long reads. Therefore, a profiling of transcript expression levels from read to transcriptome alignments was performed with NanoSim.

Error correction alters the expression profile

For 916 chromosome 1 genes with two or more transcript isoforms, where NanoSim reported a total transcript count of three or higher for the uncorrected long read data set, two measures were computed: a) the isoform consistency, measured as the cosine similarity between transcript isoform count vectors before and after correction, b) the fraction of the major isoform, i.e. the fraction

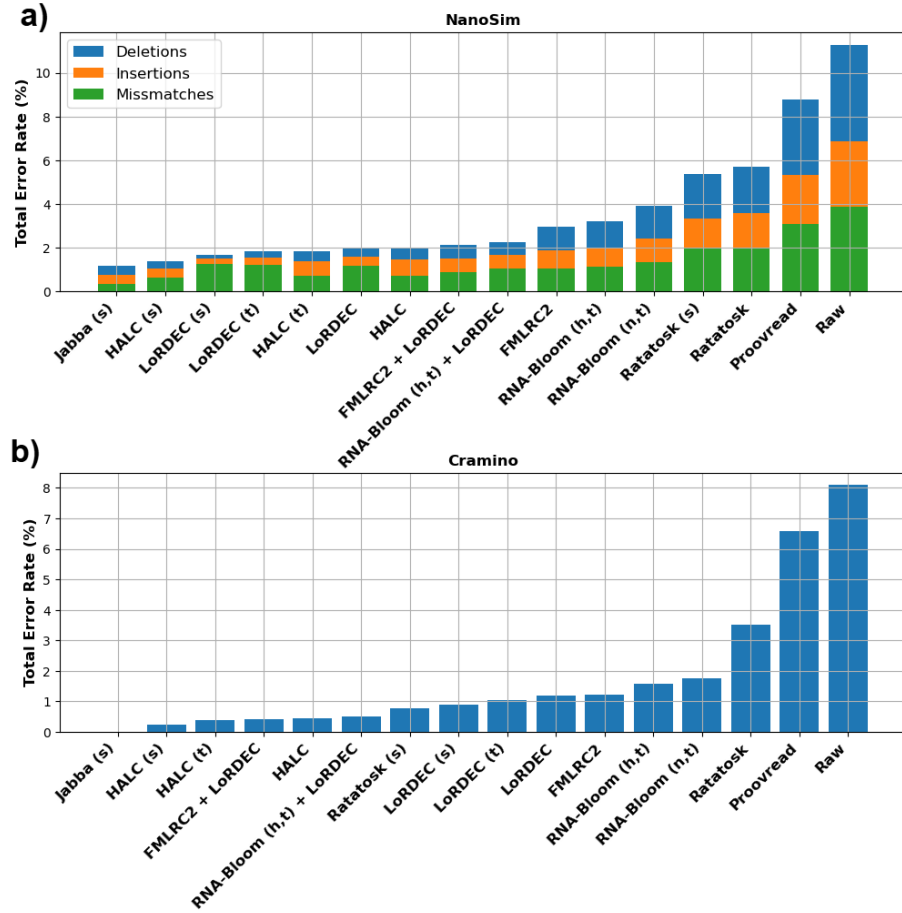


Figure 3.2: Error rates computed by a) NanoSim b) Cramino. AlignQC error rates were excluded due to their inaccuracy.
(t): trimmed long reads (with uncorrected ends removed)
(s): trimmed and split reads (with uncorrected ends and middle-regions removed)
RNA-Bloom was tested in non-hybrid (n) and hybrid mode (h)
The error rates for NanoSim, broken down by substitution, insertion and deletion, can be found in supplementary table 1.

3.2. BENCHMARK 1: HYBRID LONG READ CORRECTION

of a total counts of a gene attributed to the isoform with the highest count reported for the raw reads. Table 3.3 shows that a change of estimated transcript abundances can be observed in all corrected long read sets. In general, the expression profiles of trimmed and split reads deviate more strongly from those of raw reads than full-length reads. Contrary to this trend, RNA-Blooms trimmed reads show the highest isoform consistency.

Error correction changes the number of detectable isoforms

The change of the expression profiles from error correction is accompanied by a change of the number of detectable isoforms. Of the 6677 chromosome 1 transcripts from alternatively spliced genes, 4069 were detectable from the raw reads. All error correction tools enabled the detection of new isoforms, while also losing previously detectable RNAs. Less new RNAs are detected and more RNAs are lost when reads are trimmed and or split (see table 3.3).

Error correction reduces the fraction of the major isoform

On average, the fraction of transcript counts for each gene for the major (most expressed) isoform averages at 64.4% for the uncorrected reads (see table 3.3). For each corrected long reads the average fraction of the major isoforms is decreased.

Tool	Mean IC	Mean FM	Detected isoforms			
			total	new	lost	
Raw reads						
-	1	0.646	4069			
Full length reads						
FMLRC2	0.863	0.553 (-0.093)	3936	- 133	599	732
FMLRC2 + LoRDEC	0.878	0.587 (-0.059)	3758	- 311	536	847
HALC	0.852	0.501 (-0.145)	5547	+1478	1785	307
LoRDEC	0.895	0.586 (-0.060)	3927	- 142	491	633
Proovread	0.893	0.556 (-0.090)	4353	+ 284	757	473
Ratatosk	0.847	0.518 (-0.128)	4172	+ 103	624	521
Trimmed reads						
HALC	0.836	0.501 (-0.145)	5204	+1135	1624	489
LoRDEC	0.834	0.553 (-0.093)	3570	- 499	521	1020
RNA-Bloom (n)	0.943	0.586 (-0.060)	4177	+ 108	461	353
RNA-Bloom (h)	0.910	0.566 (-0.080)	4179	+ 110	615	505
RNA-Bloom (h) + lordec	0.900	0.596 (-0.050)	3847	- 222	479	701
Trimmed and split reads						
HALC	0.754	0.472 (-0.174)	4674	+ 605	1440	835
Jabba	0.618	0.416 (-0.230)	3147	- 922	745	1667
LoRDEC	0.768	0.502 (-0.144)	3767	- 302	783	1085
Ratatosk	0.768	0.483 (-0.163)	4773	+ 704	1364	660

Table 3.3: Summarized results of the expression profiling analysis for chromosome 1 transcripts with NanoSim.

Column 2: IC, isoform consistency (the cosine product of the vectors of transcript counts of a read set and the raw reads)

Column 3: FM, fraction of major isoform.

Column 4-6: Detectable isoforms, isoforms with transcript counts of three or higher.

Only alternatively spliced chromosome 1 genes are considered. RNA-Bloom was tested in non-hybrid (n) and hybrid mode (h)

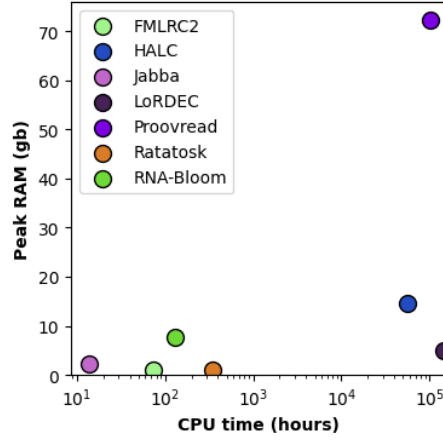


Figure 3.3: Run times and memory usage for hybrid correction of one million chromosome 1 long reads with 3.7 million chromosome 1 short reads.

3.2.3 Combining properties of correctors through serial use

Of all full length reads, those corrected by FMLRC2 showed the highest mappability and those corrected by LoRDEC had the lowest error rate according to NanoSim. By applying LoRDEC to the FMLRC2 corrected long reads, the mappability of the reads could be further increased and the NanoSim error rate reduced. According to Cramino, the resulting long reads even had the lowest error rate among the full length reads. Similarly, LoRDEC substantially improved the base error rates and also the mappability of RNA-Bloom corrected long reads. However, more isoforms were lost when LoRDEC was used in combination with FMLRC2 or RNA-Bloom than when each tool was used individually.

3.2.4 Runtimes and Memory Usages

The user time and memory usage for the hybrid long read error of the raw reads for each tool was measured with GNU time. As fig. 3.3 shows, the long read correction tools vary considerably in their running time. Jabba was, by far, the fastest tool. This can, however, in part also be attributed to the fact, that Jabba did not correct about 40% of the reads. The de Bruijn graph based correction tools required significantly less memory than the multiple sequence alignment based Proovread.

3.3. Benchmark 2: Transcriptome assembly

For a comprehensive evaluation of reference-free assembly methods, focused on hybrid methods, a total of 36 assemblies were generated *de novo*. Four short read *de novo* assemblies from 58 million real Illumina read pairs were built with RNA-Bloom, SPAdes, Trinity and Velvet+Oases. Long read assemblies were constructed from sets of five million and ten million simulated nanopore long reads with RATTLE and RNA-Bloom. Three of the previously tested hybrid long read correction methods (FMLRC2+LoRDEC, LoRDEC with trimming and RNA-Bloom) were applied to the these long reads which were then used for hybrid assemblies with IDP-denovo, RATTLE, RNA-Bloom, SPAdes and Trinity. LoRDEC was applied to the transcripts assembled by RNA-Bloom from 5 million uncorrected long reads for polishing.

In addition, for comparison purposes, reference-based assemblies were also created for all nine data

3.3. BENCHMARK 2: TRANSCRIPTOME ASSEMBLY

sets used in the *de novo* assemblies.

The Trinity assemblies with the FMLRC2+LoRDEC corrected long reads were aborted after not terminating after running for one week with 56 threads.

3.3.1 BUSCO-completeness

The BUSCO completeness of the *de novo* short read assemblies range from 83.9% (RNA-Bloom) to 92.2% (Trinity) of completely assembled BUSCOs (see fig. 3.4 and supplementary table 2). From the same data the reference-based approach achieved 93.5% (StringTie), only slightly more than SPAdes, Trinity and Velvet+Oases. The fraction of complete BUSCOs in the assemblies from IDP-denovo is low, ranging from only 1.1% to 4.6%. The BUSCO completeness for SPAdes and Trinity was only slightly increased by the use of additional long reads. RNA-Bloom achieved higher BUSCO completeness from the uncorrected long read than RATTLE. RNA-Bloom generally achieved higher BUSCO completeness with long reads than RATTLE, although the difference was smaller when hybrid-corrected long reads were used. RNA-Bloom achieved the highest proportion of complete BUSCOs for *de novo* assemblies (96.0%) and StringTie overall (98.5%), both with long reads corrected by FMLRC2 and LoRDEC. Doubling the number of long reads only slightly increased the BUSCO completeness for each assembly tool.

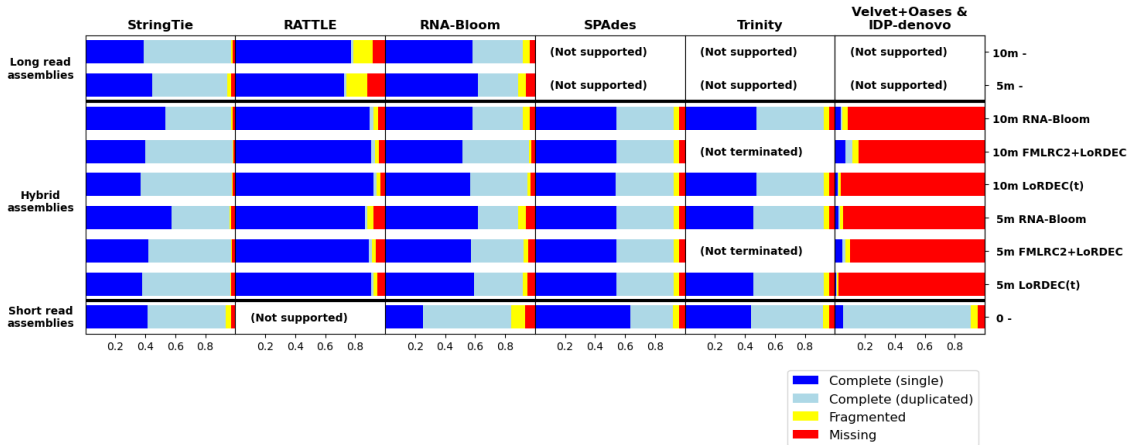


Figure 3.4: BUSCO completeness

3.3.2 rnaQUAST quality control

Non-hybrid assemblies

SR *de novo* assembly generates more and shorter transcripts

According to the annotation report from NCBI, the GRCm39 reference contains 136,032 transcripts with a mean length of 3,813 bases. Significantly more transcripts were generated by the short read *de novo* assembly tools (Trinity: 215,839; SPAdes: 309,863, Velvet+Oases: 443,769; RNA-Bloom: 509,470). On average, the assembled transcripts were rather short (SPAdes: 645; RNA-Bloom: 646; Trinity: 1,025; Velvet+Oases: 1,162). With its reference-based approach, StringTie only generated 54,291 transcripts with an average length of 3,236 from the same data. Fewer transcripts were generated by long read *de novo* assembly tools. (RATTLE with 5m LR: 21,148, RATTLE with 10m LR: 25,933, RNA-Bloom with 5m LR: 41,176, RNA-Bloom with 10m LR: 48,257). These were on average also longer (RNA-Bloom with 5m LR: 2,326, RNA-Bloom with 10m LR: 2,371, RATTLE with 5m LR: 3,543, RATTLE with 10m LR: 3,557). StringTie assembled 47,418 and

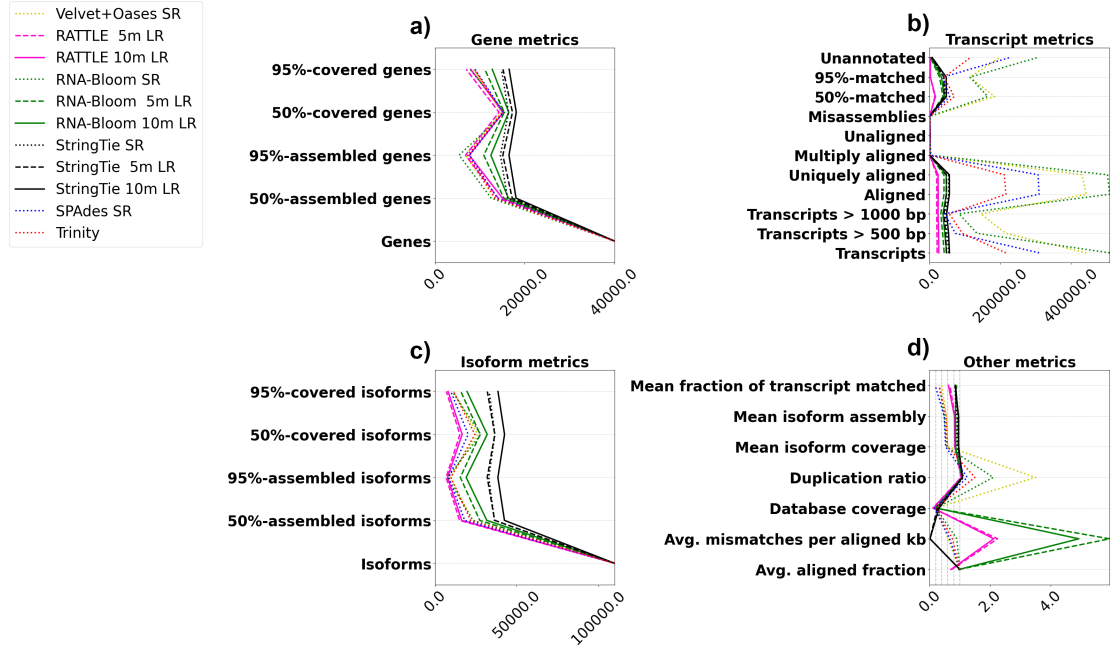


Figure 3.5: rnaQUAST results for non-hybrid assemblies

56,066 transcripts with average lengths of 2,803 and 2,949 from 5 million and 10 million long reads respectively.

LR *de novo* assembly creates more unalignable transcripts

In the previous section, we saw that some of the error-prone long reads cannot be aligned with the reference genome. As fig. 3.5 b) shows, the same is true for the transcripts assembled *de novo* from those reads (RNA-Bloom with 5m LR: 619 (1.5%), RNA-Bloom with 10m LR: 1064 (2.2%), RATTLE with 5m LR: 1,000 (4.7%), RATTLE with 10m LR: 1,746 (6.7%)). For the short read assemblies the number of transcripts that were unalignable was much lower (Trinity: 18 (<0.1%); SPAdes: 18 (<0.1%), Velvet+Oases: 93 (<0.1%); RNA-Bloom: 81 (<0.1%)).

SR *de novo* assembly shows lower mismatch rates

To avoid bias against assemblers that generate longer transcript sequences, the mismatches per transcript reported by rnaQUAST were converted into mismatches per aligned kilobase. Unsurprisingly, the reported number of mismatches per kilobase, shown in fig. 3.5 d), was lower for the short read assemblies (Velvet+Oases: 0.65; SPAdes: 0.69; Trinity: 0.88; RNA-Bloom: 0.94) than that of the long read assemblies (RATTLE with 5m LR: 1.97, RATTLE with 10m LR: 1.92, RNA-Bloom with 5m LR: 5.3, RNA-Bloom with 10m LR: 4.5). Doubling the long read coverage had less of an impact on the mismatch rate for RATTLE, which by default requires a minimum coverage of 5, compared to that of RNA-Bloom, which only requires a minimum coverage of 3 by default.

RNA-Bloom produces the fewest misassembled transcripts

Transcripts whose parts have been aligned with locations in the genome that are distant or on different strands are considered as misassemblies. 0.16%-0.74% of transcripts assembled *de novo* are misassemblies (see fig. 3.5 b)). (SPAdes: 496 (0.16%); Trinity: 498 (0.23%); RNA-Bloom: 804 (0.16%); Velvet+Oases: 3383 (0.74%)). While RNA-Bloom created less misassemblies (RNA-Bloom with 5m LR: 16 (0.05%), RNA-Bloom with 10m LR: 35 (0.07%)), RATTLE produced

3.3. BENCHMARK 2: TRANSCRIPTOME ASSEMBLY

higher fractions of misassemblies (RATTLE with 5m LR: 265 (1.25%), RATTLE with 10m LR: 305 (1.18%))

SR *de novo* assembly contain many unannotated transcripts

The short read *de novo* assembly tools created many isoforms that are not included in the reference annotation (Trinity: 115,098/ (53.4%); SPAdes: 226,613 (73.1%); Velvet+Oases: 197,747 (44.6%); RNA-Bloom: 305,154 (60.0%)). StringTie output only 5404 (10.0%) unannotated transcripts from the same data. The long read *de novo* assemblers RATTLE and RNA-Bloom assembled significantly less unannotated transcripts (RNA-Bloom: 2,286 (5.4%) from 5m reads and 4,219 (8.6%) from 10m reads, RATTLE: 2,051 (7.9%) from 5m reads and 1128.0 (5.3%) from 10m reads). With 5,496 (11.6%) unannotated transcripts from 5m long reads and 7,293 (13.0%), StringTie assembled similar amounts of unannotated transcripts.

Comparison of hybrid and non-hybrid methods

IDP-denovo deteriorates assembly quality

IDP-denovo produced assemblies with duplication ratios that were even higher than those of the input short read assembly from Velvet+Oases (4.5-11.0 compared to 3.5) while covering significantly less genes and isoforms. Mismatch rates per kilobase range from 15.4 to 26.0, multiple times higher than for any other assembly tool.

SR assemblies are largely unaffected by additional long reads

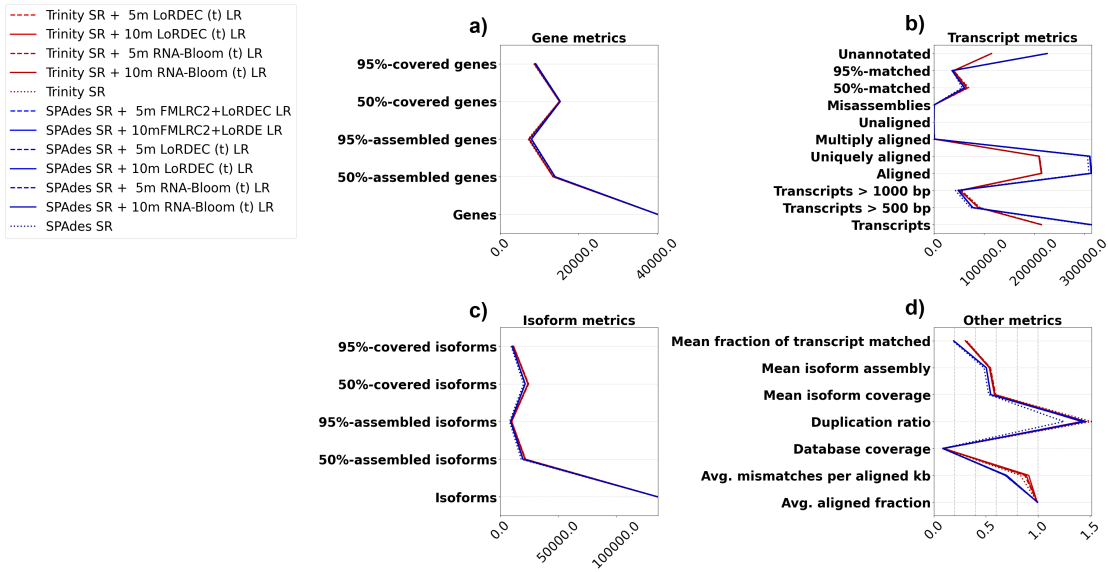


Figure 3.6: rnaQUAST results for hybrid SPAdes and Trinity assemblies.

Hybrid LR correction and transcript polishing decrease mismatch rates in LR *de novo* assemblies

Figure 3.7 and fig. 3.8 the rnaQUAST results for RATTLE and RNA-Bloom transcripts assembled from 5 million raw and corrected long reads. Most striking is the effect of hybrid error correction on the number of mismatch per kilobase of the assembled transcripts. Using five million hybrid corrected reads, RATTLE achieved mismatch rates ranging from 1.10 (FMLRC2+LoRDEC) to 1.56 (RNA-Bloom) compared to 1.97 with uncorrected reads (fig. 3.7 d)). Mismatches rates for

RNA-Bloom with hybrid corrected reads range from 0.99 (LoRDEC with trimming) to 3.19 (RNA-Bloom) compared to 5.3 with raw reads (fig. 3.8 d)). RATTLE also shows increased mean isoform coverage and assembly with hybrid corrected long reads (fig. 3.7 a) and c)). For RNA-Bloom these were slightly reduced with hybrid corrected long reads (fig. 3.8 a) and c)). Applying LoRDEC for post-assembly hybrid polishing of the transcripts assembled from 5 million uncorrected long reads with RNA-Bloom reduced the mismatch rates from 5.3 to 1.8 without decreasing the isoform coverage and assembly (see fig. 3.9).

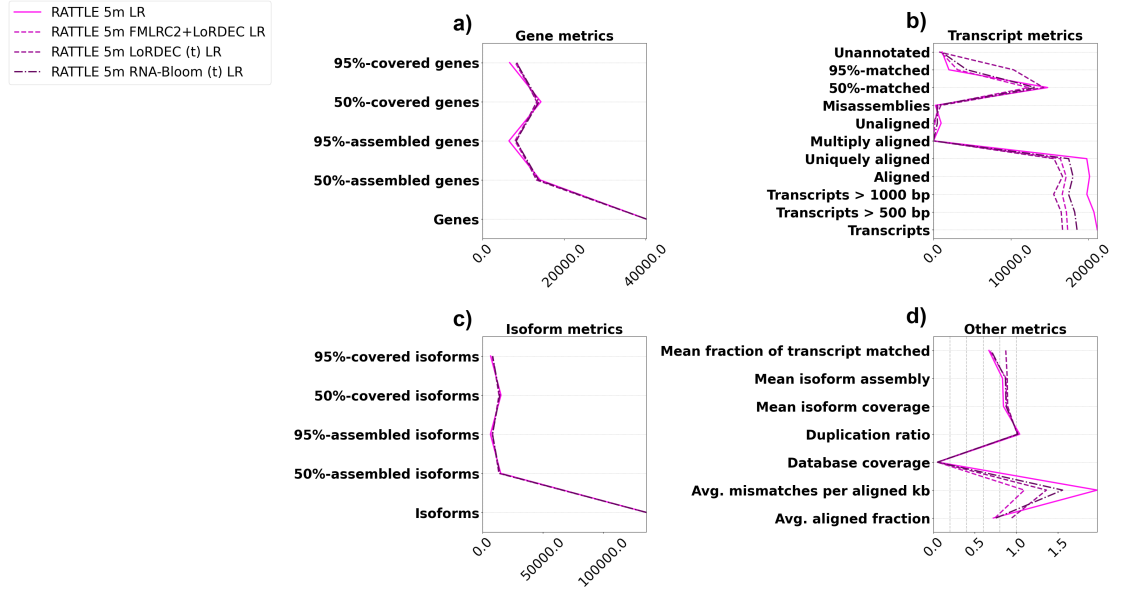


Figure 3.7: rnaQUAST results for RATTLE transcripts assembled from 5 million raw and error corrected long reads for non-hybrid assemblies

3.3.3 Runtimes and Memory Usages

The user time and memory usage for the assembly from RNA-Bloom corrected long reads was measured with GNU time. As fig. 3.10 shows, the assembly tools vary considerably both in their running time and required memory. SPAdes was by far the fastest tool, while RATTLE was the slowest. For tools that construct a de Bruijn graph from Illumina short reads, i.e. Velvet+Oases with IDP-denovo, SPAdes and Trinity, the required memory did not change with increasing long-read coverage. For the long-read assemblers RATTLE and RNA-Bloom, on the other hand, doubling the long-read coverage leads to approximately double the required memory. Although the iterations of long read error corrections to be performed for RNA-Bloom were set to 0, it still generates a de Bruijn graph for error correction. In this comparison, this tool is therefore at a disadvantage. Nevertheless, it requires fewer resources than RATTLE and was still the second fastest tool, at least for the dataset with 10 million long reads.

3.3. BENCHMARK 2: TRANSCRIPTOME ASSEMBLY

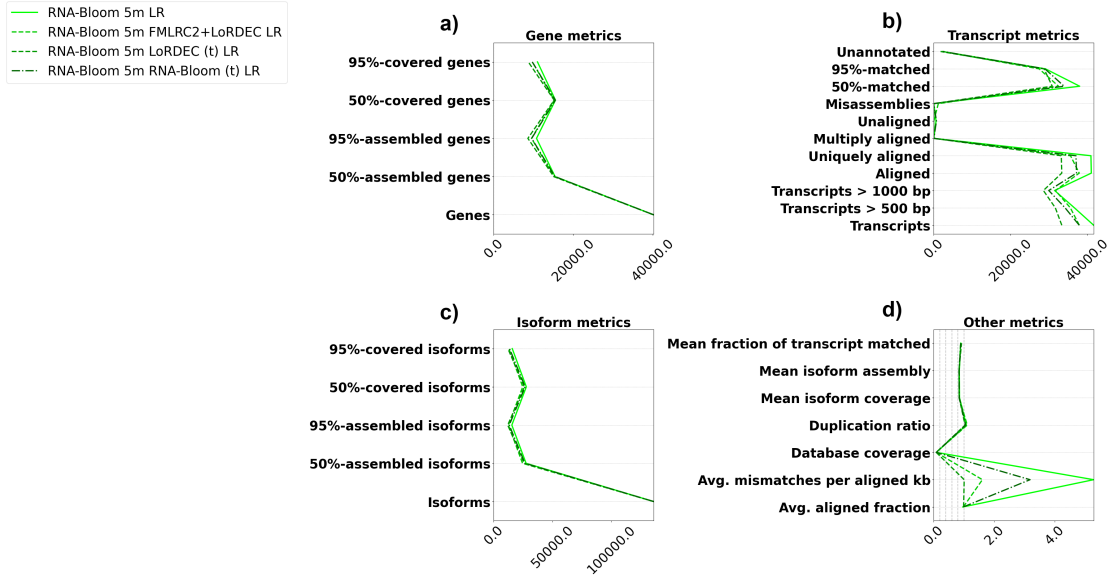


Figure 3.8: rnaQUAST results for RNA-Bloom transcripts assembled from 5 million raw and error corrected long reads for non-hybrid assemblies

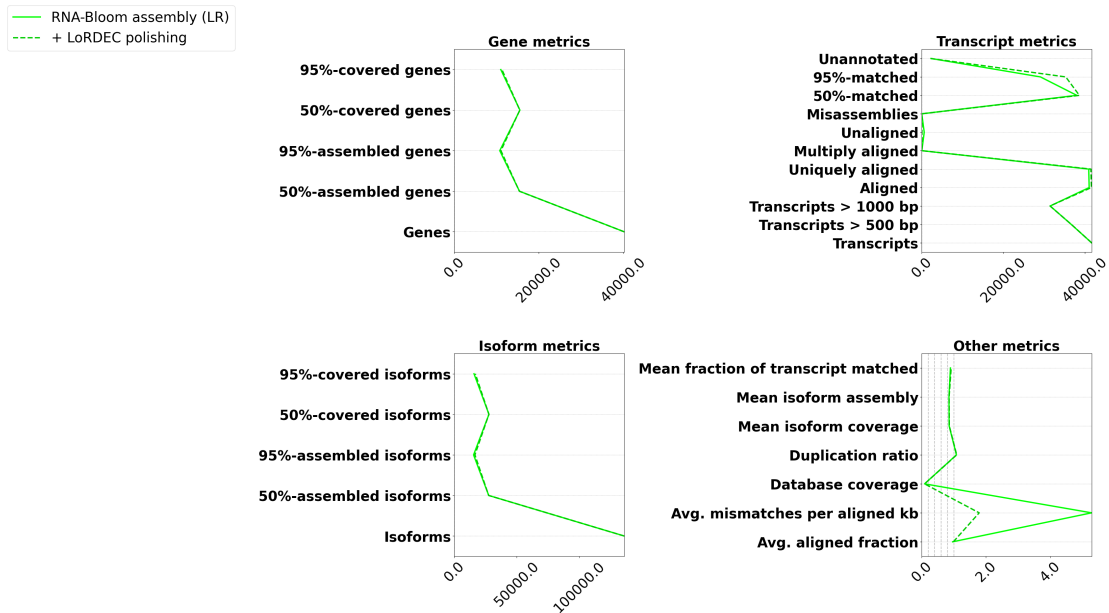


Figure 3.9: rnaQUAST results for RNA-Bloom transcripts assembled from 5 million raw before and after transcript polishing with LoRDEC

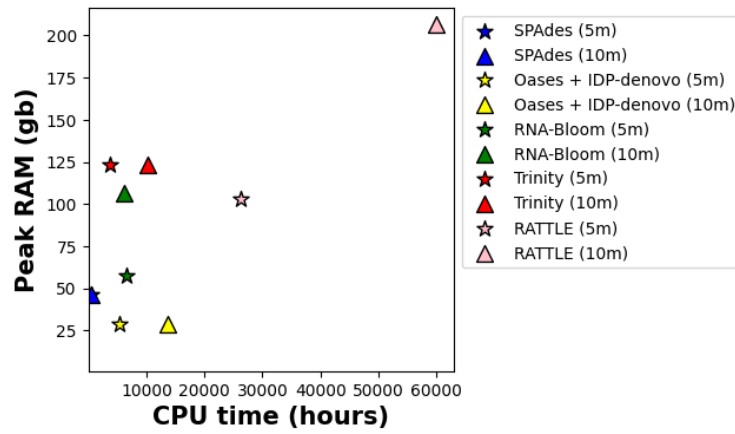


Figure 3.10: Run times and memory usage for hybrid assemblies constructed with RNA-Bloom corrected long reads

3.4. *Proteus anguinus* transcriptome

The findings from previous benchmarks were used to create an improved transcriptome assembly for the long-lived olm.

3.4.1 Assembly with improved continuity

The sequencing and preprocessing resulted in a total of 253,408,682 Illumina short read pairs and 8,301,909 MinION long reads. RNA-Bloom assembled 260,381 transcripts from the long reads. 16,564,668 (6.5%) Illumina read pairs were identified as potential human or microbial contaminations and removed with Kraken2. The assembled transcripts were then polished using LorDEC and the filtered Illumina reads. 11,525 (4.426%) of the transcripts that LorDEC was unable to polish, were removed as they are likely to also stem from contamination or are of low quality. From the preliminary Trinity assembly published in our publication, 4465 and 362 protein coding transcripts giving the best hits with DIAMOND for proteins in the UniProt Knowledgebase and with the BUSCO tool for vertebrate BUSCOs in the OthoDB v10, respectively, were added to the RNA-Bloom assembly.

The new RNA-Bloom long read assembly contains only about half as many transcripts as the old Trinity assembly (see table 3.4). However, on average, these are about a quarter longer and more of them contain complete open reading frames (ORFs). Nevertheless, the RNA-Bloom assembly has a significantly reduced BUSCO-completeness compared to the Trinity assembly (See fig. 3.11). For 4,827 proteins for which corresponding transcripts could be identified with first DIAMOND and then BUSCO in the old assembly but not in the new assembly, the transcripts with the best scores were added to the new assembly. While transcripts for 795 of the total 3,354 vertebrate BUSCOs are missing in the transcriptome assembled with RNA-Bloom, only 121 are missing after the addition of transcripts from the old assembly in which 126 are missing. However, with 424 compared to 94, the number of fragmented BUSCO transcripts in the new assembly is still higher.

Assembly	# Transcripts	# ORF	Lengths			# Bases	N90	N50
			min	mean	max		N90	N50
Irrisarri et al.	74,834	13,707	192	542	12,217	40557,227	268	634
Holtze et al.	540,179	76,482	200	756	29,486	408,500,084	285	1,405
RNA-Bloom	248,856	89,454	71	1,026	36,795	257,120,836	480	1,592
RNA-Bloom +	253,321	93,067	71	1,055	36,795	269,179,055	490	1,658
RNA-Bloom ++	253,683	93,428	71	1,060	36,795	270,890,139	492	1,670

Table 3.4: Basic statistics on olm transcriptomes reported by TransRate.

Irrisarri et al., 2021 [9]

Holtze et al., 2025 [10]

RNA-Bloom : LorDEC polished and contamination filtered RNA-Bloom assembly.

RNA-Bloom + : RNA-Bloom supplemented with missing protein coding transcripts from Holtze et al. identified with DIAMOND and UniProt Knowledgebase

RNA-Bloom ++ : RNA-Bloom + supplemented with missing vertebrate BUSCO transcripts from Holtze et al. identified with the BUSCO tool and OrthoDB v10

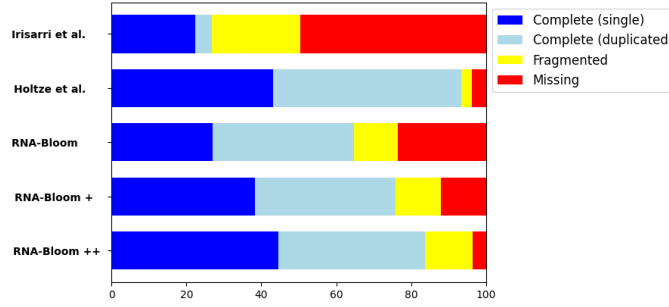


Figure 3.11: Completeness of vertebrate BUSCOs in olm transcriptomes.

Irrisarri et al., 2021 [9]

Holtze et al., 2025 [10]

RNA-Bloom : LorDEC polished and contamination filtered RNA-Bloom assembly.

RNA-Bloom + : RNA-Bloom supplemented with missing protein coding transcripts from Holtze et al. identified with DIAMOND and UniProt Knowledgebase

RNA-Bloom ++ : RNA-Bloom + supplemented with missing vertebrate BUSCO transcripts from Holtze et al. identified with the BUSCO tool and OrthoDB v10

3.4.2 p53 characterization reveals potential phosphorylation targets

The p53 plays a crucial role in tissue homeostasis and aging [12]. The amino acid sequence of the p53 protein of the olm, whose transcriptome was assembled here, differs in four positions from those in a previously published protein sequence of a different olm published by Irrisarri et al. [9]. Olm populations living in cave systems that have been separated from each other for a long time may exhibit a greater evolutionary distance between them. The different amino acids are located in disordered stretches of the protein and do not affect its structure.

A multiple sequence alignment for p53 sequences of different amphibians shows two amino acid insertions for *Proteus anguinus* (olm) and its shorter-lived relative *Ambystoma mexicanum* (axolotl). p53 includes an additional serine at the insertion site for both species (see fig. 3.12). A substitution of threonine for another serine can only be found in the olms p53. The additional serines of the olm are located in loops within the DNA-binding domain (see fig. 3.13). Their accessibility makes them potential targets for phosphorylation.



Figure 3.12: Multiple protein alignments of the p53 dimerization region for *Proteus anguinus*, (olm) *Ambystoma mexicanum* (axolotl) and other amphibians. The source of the protein sequences and the trivial names of each species are described in section 2.4.7.

A comparison of the predicted structures of p53 for the olm, the axolotl and the african clawed frog shows differences in the C-terminal oligomerization domain (see fig. 3.14). For the frog, it contains a single helix, while for the axolotl and the olm, it breaks down into two separate helices. One of them appears to be shortened in the olm.

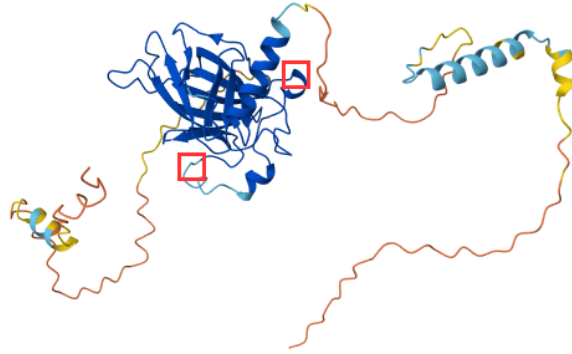


Figure 3.13: Protein structure of p53 in *Proteus anguinus* with two additional serines within the DNA-binding domain (highlighted by the red squares).
Bottom left: insertion
Top right: substitution

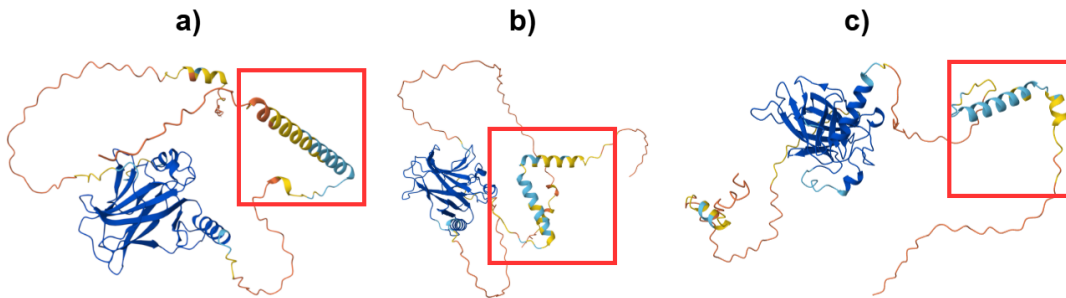


Figure 3.14: Protein structure of p53 predicted with AlphaFold3 for a) *Xenopus laevis* (african clawed frog) b) *Ambystoma mexicanum* (axolotl) and c) *Proteus anguinus* (olm). The structure of the C-terminal oligomerization domain (highlighted by the red squares) differs across all three species.

4. Discussion

This work provides insights into the practical use of Illumina short and nanopore long reads in the context of hybrid *de novo* transcriptome assembly from pooled tissue samples and applies the findings to the transcriptome of the olm.

4.1. Benchmark: Long read error correction

In section 3.2 a multitude of measures for the quality of the corrected and uncorrected long reads were presented. None of the tools tested was able to outperform all others across all metrics. The choice of tool therefore requires prioritisation of the various metrics. In the context of transcriptomic long read error correction, it is important that individual base segments are correctly replaced without changing the basic structure, i.e. the exon composition, of a long read. The former is represented mostly by the base error rate, the latter by the isoform consistency.

4.1.1 Hybrid long read error correction improves mappability and base error rates

First, the quality of the long reads was evaluated based on alignments with the reference genome. The rate at which erroneous substitutions, insertions and deletions occur in within those long reads is particularly meaningful. For individual tools with full length read output, a reduction of this base error rate from 11.29% to as little as 1.96% (LoRDEC) according to NanoSim or from 8.09% to 0.45% (HALC) according to Cramino can be observed.

In fact, three different error rates were calculated for each set of long reads, using AlignQC, Cramino and NanoSim respectively. The rate calculated by AlignQC is based on a sample of long read alignments with at least one million bases. Depending on the average read length those samples might only include a few hundred long reads, and is not necessarily representative due to its small size. In fact it is possible to change the reported error rate by using a different number of threads for the computation of the alignments. The error rates of Cramino and NanoSim take into account the alignments of all long reads and are therefore more accurate. Cramino considers insertions and deletions of multiple consecutive bases as single errors. NanoSim, on the other hand, counts each inserted or deleted base as a separate error. Therefore, its error rates better reflect the actual difference between the true sequence and the read. However, it can also be argued that substitutions should be penalised more heavily than insertions and deletions, even if they are two bases long rather than just one. Insertions and deletions that would cause a frameshift and thereby a disruption of the open reading frame and likely also protein function can be corrected using alignments to proteins in a reference database. [64]

Ratatosk, which was used to correct the long reads for our preliminary Trinity assembly output reads with comparably high error rates of 5.70% according to NanoSim and 3.53% according to Cramino. The higher error rates for Ratatosk should not be interpreted as a general statement about

its quality. Ratatosk has been specialized for the correction of genomic long reads in the context of variant calling. This specialisation is reflected in several steps of its algorithm [17] and appears to render the tool unsuitable for the correction of transcriptomic long reads.

The base error rates, however, only represent regions of the long reads that are of high enough quality to be aligned in the first place. Subsequently, the fraction of aligned reads and bases must also be taken into account. 84.5% of the bases in the raw reads were aligned. All tools were able to improve the mappability, with FMLRC2 achieving the highest fraction of aligned bases for full length reads with 90.6%. HALC and LoRDEC have similar NanoSim base error and mapping rates.

4.1.2 Full-length vs trimmed and split reads: A trade-off between continuity and base accuracy

The error rate can be further reduced and fraction of aligned bases increased by removing uncorrected regions at the ends of the reads (trimming) or in the middle of the reads (splitting). However, this comes at the expense of read continuity. It is reasonable to assume that the splitting of long reads would also lead to a fragmentation of the resulting assembly and should therefore be discouraged. Additionally, assemblers may still be able to achieve decent error rates in regions where reads were not corrected by forming consensus sequences from multiple long reads.

4.1.3 Long read correction changes expression profiles

One effect of long read correction, first described by Lima et al, is that the isoforms of the reads can change and be lost entirely. [14] The isoform of a long read may change if it is not aligned with its corresponding path in the de Bruijn graph. The correct path may not be found due to a particularly high number of sequencing errors within a region of the long read, or it might not be present in the graph due to insufficient short read coverage. The latter is consistent with the observation by Lima et al. that low expressed isoforms are more likely to be lost.

To measure the capability of a correction method to preserve isoforms and expression profiles for alternatively spliced genes, the isoform consistency defined as the cosine product of the vectors of the isoforms expression values was used. The expression values are calculated from alignments of the long reads and the reference transcript sequences. It can be assumed that the mapping rates here are higher than those mentioned above, as these are not spliced alignments. The expression values can therefore be considered to be reasonably reliable.

The highest isoform consistency was reported for RNA-Bloom, which is the only tool that uses an expression-level aware approach to long read correction. The reported values were 0.910 when the correction was performed with short reads and 0.943 when performed without. This suggests that the typically less uniform coverage in Illumina data hinders RNA-Bloom's expression-level aware approach. This effect may increase for real nanopore and Illumina RNAseq data. In this benchmark, I used simulated long reads whose measured expression profile correlates very strongly with that of the corresponding Illumina short reads (Spearman/Pearson ≈ 0.99 , see fig. 2.1). Such strong correlations cannot always be found for real data sets. [70] The higher the reported isoform consistency for a correction method, the more complete the RNA-Bloom assembly constructed from long reads corrected with the method was.

4.1.4 Different properties of LR correctors can be combined through serial use

Previous benchmarks of long read correction software tools tested them exclusively individually. [11, 12, 13, 14] In this paper, I show that the properties of different methods can be combined

by using them in series. Of all full length reads, those corrected by FMLRC2 showed the highest mappability and those corrected by LoRDEC had the lowest error rate according to NanoSim. By applying LoRDEC to the FMLRC2 corrected long reads, By applying LoRDEC to the FMLRC2, the mappability of the reads could be further increased and the NanoSim error rate reduced. According to Cramino, the resulting long reads even had the lowest error rate among the full length reads. Similarly, LoRDEC substantially improved the base error rates in RNA-Bloom corrected long reads. However, more isoforms were lost when LoRDEC was used in combination with FMLRC2 or RNA-Bloom than when each tool was used individually.

4.1.5 Comparison with the results of a similar study by Lima et al.

Lima et al. already performed a similar but more detailed benchmark. However, they used RNA sequencing data of only a single tissue, and evaluated a different set of correction tools. The increased isoform diversity in RNA sequencing data pooled from multiple tissues poses a more difficult challenge for long read correction tools.

The error rates for HALC are similar in both studies. However, the error rates for LoRDEC and Proovread are significantly lower and higher, respectively. In their paper, Lima et al. found that the correction of transcriptomic long reads changed the measured expression profile. In particular, they observed a loss of isoforms and the correction of long reads towards the major (most expressed) isoform. [14] In part, these observations also apply to the findings of this thesis. In this benchmark, individual isoforms were also lost due to the correction of long reads with all tools. However, with HALC, Proovread, Ratatosk and RNA-Bloom, it was even possible to detect more isoforms overall after the correction. Also, instead of an increase of the fraction of the major isoform I found a reduction of this fraction. The differing results are not necessarily contradictory, but rather reflect the differences in the underlying data. Due to the higher isoform diversity in the data used here, there are more possibilities for changing the structure of a read of the major isoform towards a minor isoform. Conversely, it is less likely that a read of a minor isoform will be changed to the major isoform instead of another minor isoform.

4.2. Benchmark: Transcriptome *de novo* Assembly

4.2.1 Reference-based assemblies for comparison

Given a high quality genome assembly is available, reference-based assembly approaches have the potential to yield better results than *de novo* assembly approaches. In order to identify pairs of overlapping sequence reads, *de novo* assembly can rely only on the information of the overlapping parts of pairs of reads. Furthermore, distinguishing true biological partial read overlaps from short random sequence similarities with with reasonable confidence requires a minimal overlap length. Reference-based assembly identifies overlaps from the positions of read to genome alignments. Consequently, it can employ the full read information to identify overlaps without the need for a lower limit of the overlap length. In fact, this benchmark also shows, that StringTie creates more complete assemblies with less structural errors. For each data set, they can be considered as reference solution.

4.2.2 Short read vs long read assembly: A trade-off between base accuracy and structural correctness

Compared to long reads, short reads can be used to assemble transcripts with significantly lower mismatch rates *de novo*. At the same time, these are less complete and more often unannotated. The study design may lead to a bias against short read assemblers. While the simulated long reads originate exclusively from annotated transcripts, it is possible that the real short reads also

originate from real but unannotated mouse transcripts. The expected relative completeness of the reference annotation and the fact that StringTie generates significantly fewer unannotated transcripts with the same data indicates that the vast majority of the unannotated transcripts of the *de novo* assemblers are, however, assembly errors.

The decision to use either short reads or long reads for a *de novo* transcriptome assembly is therefore one between base-level accuracy on the one hand and structural correctness and continuity on the other.

4.2.3 Increasing base-level accuracy in LR assemblies using short reads

IDP-denovo uses long reads to extend transcripts assembled from short read, while SPAdes and Trinity both utilize long reads during short read assemblies to navigate their respective assembly graphs. Neither approach, however, leads to an improvement of the output assemblies. On the other hand, using short reads pre-assembly in hybrid long read error correction or post-assembly in hybrid transcript polishing increases the base-level accuracy of long read assemblies. RATTLE assemblies benefit most from hybrid long read error correction. Not only does it reduce the number of base errors in the assembled transcripts, but it also enables RATTLE to assemble some of them in the first place. RNA-Bloom, on the other hand, achieves the highest completeness when it does not use hybrid-corrected long reads. As discussed in section 4.1.3, the correction of long reads may change their isoforms. This is least prevalent when the expression-level aware correction approach of RNA-Bloom is used, provided it is not hindered by additional short reads. Compared to the base accuracy of the input long reads that of the transcripts assembled by RNA-Bloom is already improved significantly. This allows tools such as LoRDEC, which do not follow an expression-level conscious approach, to identify the paths corresponding to the transcripts in their de Bruijn graph more reliably. Post-assembly hybrid transcript polishing is therefore the best method for combining the advantages of RNA-Bloom in terms of continuity and structural correctness of the assembled transcripts with the base level accuracy of short reads. Another minor advantage of transcript polishing with LoRDEC is that bases in corrected regions are marked with capital letters, while bases in uncorrected regions are marked with lowercase letters. This allows at least a rough distinction to be made between more and less reliable transcript regions.

4.2.4 Comparison of RATTLE and RNA-Bloom

The assemblies produced by RATTLE are less complete than those produced by RNA-Bloom. This can largely be explained by the higher minimum coverage requirement (five for RATTLE, three for RNA-Bloom). However, the increase in transcriptome completeness according to BUSCO and rnaQUAST suggests that RATTLE has difficulty correctly associating long reads with high error rates during clustering.

4.2.5 Reliability of completeness estimation with BUSCO

The BUSCO completeness scores for RATTLE and RNA-Bloom assemblies constructed from hybrid corrected long reads was similar for both tools. The overall completeness, however, differed more significantly according to rnaQUAST. BUSCO completeness therefore only provides a rough estimate of the overall completeness of an assembly.

4.3. Making biological statements about *de novo* assembled transcripts

The integrated use of Illumina short reads and Nanopore long reads elevates *de novo* transcriptome assembly to a new level of accuracy. Nevertheless, even then at least one in twenty assembled transcripts was reported as unannotated by rnaQUAST, i.e. containing structural errors. Furthermore, each assembled transcript contains on average at least one error at the base level.

By comparing assembled genomic or transcriptomic sequences with those of related organisms, researchers attempt to draw conclusions on molecular biological characteristics special to the newly examined organism. Given the aforementioned error rates, it is to be expected that many sequence differences do not originate from evolution but rather from sequencing and assembly errors. It is therefore important to be critical of the assembled transcript sequences and, if necessary, to check in individual cases whether they are well supported by the read data.

4.4. Future work in the field of *de novo* transcriptome assembly

Currently, there is no approach that is fully focused on hybrid *de novo* transcriptome assembly. Trinity and SPAdes were originally designed as short read assemblers. Only later have they been extended to integrate long reads, however, with little success as this benchmark shows. RATTLE assemblies are based purely on long reads and requires the use of additional tools for hybrid long read error correction. RNA-Bloom on the other hand is capable of performing hybrid long read error correction on its own. RNA-Bloom, on the other hand, can perform hybrid long read correction itself, but does not take into account the special characteristics of short reads. It is therefore also best used in combination with additional tools. A tool that is fully focused on hybrid assemblies and the specific characteristics of its input read data could lead to even better results. However, with the increasing prevalence of HiC-Pacbio and R2C2 reads, long reads with ever-improving quality, it is questionable whether the need for such a tool will last.

Even though sequencing methods and algorithms for transcriptome assemblies are constantly evolving, assembly errors cannot be ruled out. It would therefore be necessary not only to improve the results themselves, but above all to create and establish a framework that promotes a more critical approach to them. Currently, assembled transcript sequences are output in simple FASTA format, without any indication of how well their structure or individual bases are supported by the read data. A tool that pursues a similar goal is TransRate. However, it is an older project that is no longer maintained and cannot yet include long reads. Furthermore it only provides scores for transcripts but not for their individual bases. citeTransRate The quality of individual bases could, for example, could be estimated on the basis of the reads that can be aligned with the assembled transcripts and specified using the FASTQ format.

4.5. *Proteus anguinus* transcriptome

4.5.1 Assembly

Despite the higher number of fragmented BUSCOs, the transcriptome assembly created in this work certainly represents an improvement over the previous assembly.

For the previous assembly, long reads were corrected with Ratatosk, which was designed for the correction of genomic long reads in the context of variant calling. However, the benchmark of hybrid long read correction tools showed, that is not well suited for the correction of transcriptomic long

reads from pooled tissue samples. Furthermore, the benchmark from Assembly tools shows that the results from Trinity are hardly influenced by additional long reads anyway. Trinity assembled far too many and rather short transcripts for the test data, about half of which contained structural errors. In the benchmark, RNA-Bloom assemblies from long reads had higher continuity and completeness but less structural errors. In part, these findings seem to also apply to the RNA-Bloom assembly for the olm. It contains less but longer transcripts and more open reading frames. However, its BUSCO completeness is reduced compared to that of the Trinity assembly. While the number of long reads for the olm is similar to the number of long reads in the mouse benchmark, the number of short reads for the olm is significantly higher. After supplementing transcripts from the older Trinity assembly, it can be assumed that the final transcriptome assembly will contain at least one high-quality isoform for most protein-coding genes. Furthermore, transcripts presumed to originate from foreign organisms were indirectly filtered using Kraken2 and LoRDEC.

4.5.2 p53 characterization reveals potential phosphorylation targets

A comparison of the aminoacid sequence of the tumor suppressing protein p53 of the olm and six different amphibians, revealed that the olms p53 has two inserted serines located within its DNA binding domain. One of the insertions is also found in the p53 of the axolotl, while the other appears to be specific to the olm. Since they are both located in accessible loops, these insertions might be potential targets for post-translational modifications in the form of phosphorylation. Furthermore, a comparison of the structure of the olms p53 with that of the axolotl and the african clawed frog revealed potential differences in the structure of the C-terminal oligomerization domain of p53 which might also impact the function of p53 in the olm.

A previously published sequence of p53 for the olm differs in a total of four amino acids from the sequence determined here. However, the differences do not affect the protein structure. Upon enquiry, it was discovered that the previously published sequence originated from the transcriptome of an olm from a research programme at SETE Moulis CNRS. This programme began in the 1950s with olms collected in Slovenia. [71] The cave salamander whose transcriptome was assembled here, on the other hand, originates from a cave near Split in southern Croatia, about 300km away from Slovenia.

4.5.3 Future work on the olm

The cave salamander shows high resilience to anoxic conditions and can last for several hours without oxygen. The cave salamander shows remarkable resilience to anoxic conditions. Other species also experience increased oxidative stress during subsequent reoxygenation, which they counteract with increased activity of antioxidant enzymes. This increased enzyme activity is only partially observed in the olm, yet it does not appear to experience increased oxidative stress. These findings come from a study by Issartel et al., 2017, where they also suggests alternative ways in which the olm avoids oxidative stress. [6] The molecular mechanisms that allow the olm to efficiently avoid oxidative stress during reoxygenation might also increase its genomic stability. The new transcriptome assembly could help to investigate them.

5. Conclusion

This thesis investigated methods of *de novo* transcriptome assembly from short, accurate and long error-prone sequence reads. Assemblies using either short or long reads reflected the characteristics of the input data. Transcripts assembled with short reads have high accuracy at the nucleotide base level but are often incomplete or contain structural errors. In contrast, transcripts assembled with long reads are more complete and contain fewer structural errors. However, they contain significantly more nucleotide base errors. Three categories of hybrid approaches attempt to improve *de novo* transcriptome assembly by combining the advantages of both types of reads. The first, implemented only in IDP-denovo, uses long reads to extend incomplete transcripts assembled from short reads. The second, implemented in SPAdes and Trinity, aligns long reads with assembly graphs constructed from short reads in order to help identifying paths corresponding to transcripts. However, neither approach led to any noticeable improvement compared their respective non-hybrid short read assemblies. The third combines long read assembly with either hybrid long read correction pre assembly or transcript polishing post assembly. The majority of correction tools also align long reads with assembly graphs constructed from short reads. The paths in the assembly graph dictates the correction of the corresponding long reads. Hybrid error correction significantly reduces the number of base errors in long reads, however, when long reads are not aligned correctly, their isoform may change during the correction. A special correction module is included in the long read assembler RNA-Bloom. On the one hand, it constructs the graph used for correction from the long reads themselves; short reads can be added optionally. On the other hand, it is the only tool that takes an expression-level conscious approach when aligning the long reads with the assembly graph. This allows it to preserve the isoforms of the reads better than other tools, especially when used without additional short reads. The transcripts assembled by RNA-Bloom already exhibit a significantly reduced base error rate compared to the uncorrected long reads. This means that they can also be corrected by other hybrid correction tools, such as LoRDEC, without changing their structure. The best strategy for hybrid *de novo* transcriptome is the self-correction of and assembly from long reads with RNA-Bloom followed by a hybrid transcript polishing.

These findings were then applied to create an improved transcriptome assembly for the long-lived amphibian *Proteus anguinus* (olm). The new transcriptome assembly may help to explore the molecular mechanisms that give the olm its longevity. Some interesting changes to the sequence and structure of the tumor suppressing protein p53 of the olm were already described in this thesis. Next, the olms ability to avoid oxidative stress could be further investigated.

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Acronyms

DBG de Bruijn graph. 3, 4, 6–9, 42

fl full length long reads. 8, 9, 42

LR Long Read. 9, 26–28, 42

OLC Overlap Layout Consensus. 4, 6, 7, 42

p53 tumor suppressor protein 53. 32–34, 40, 42

s trimmed and split long reads. 8, 9, 42

SMRT Single-Molecule-Real-Time. 7, 42

SNP single nucleotide polymorphism. 42

SR Short Read. 7, 9, 26–28, 42

t trimmed long reads. 8, 9, 42

Appendix

.1. Tools and Versions

Tool	Version
Correction	
FMLRC2	0.1.7
HALC	latest commit (Jun 11, 2018)
LoRDEC	0.9.0
Proovread	2.14.0
Ratatosk	0.9.0
RNA-Bloom	2.0.1
Assembly	
IDP-denovo	2
Oases	0.2.09
RATTLE	1.0
RNA-Bloom	2.0.1
SPAdes	4.0.0
StringTie	2.2.1
Trinity	2.15.1
Others	
BUSCO	5.6.1
eggNOG-mapper	2.1.12
FastQC	0.12.1
Kallisto	0.48.0
Kraken2	2.1.3
MAFFT	7
MapSampler	1.0
NanoSim	3.2.2
Minimap2	2.29-r1283
Porechop	0.2.4
rnaQUAST	2.2.4
Samtools	samtools 1.18
SSfSBT	
TransDecoder	5.5.0

Table 1: Used tools and their versions

Reads	AlignQC error rates	
	128 threads	1 thread
Raw	10.51	10.62
FMLRC2	0.98	1.21
FMLRC2 + LoRDEC	0.42	0.39
HALC	0.39	0.38
LoRDEC	0.90	0.83
Proovread	7.86	?
Ratatosk	3.11	2.97
HALC (t)	0.39	0.28
LoRDEC (t)	0.74	0.75
RNA-Bloom (t)	1.56	1.60
RNA-Bloom (t) + LoRDEC"	0.36	0.33
HALC (s)	0.19	?
Jabba (s)	0.01	0.01
LoRDEC (s)	0.50	0.62
Ratatosk (s)	0.55	?

Table 2: AlignQC error rates can change by performing alignment with different numbers of threads:

Long reads were aligned with the genomes using Minimap2 with 128 and 1 thread, respectively. Missing values were the analysis was not repeated with single-threaded Minimap2 are marked with “?”

.2. Alternative AlignQC error rates

The error rates shown in the results section are based on alignments where Minimap2 ran with 128 parallel threads. For the first 5,000 long reads in each case, I repeated the analysis for some of the data sets, but ran Minimap2 with only a single thread. Using a different number of threads changes the order in which alignments are output by Minimap2. As a result, AlignQC uses different samples for the evaluation of the base error rates. Table 2 shows, that the changed samples affect the error rates significantly.

AlignQC samples the best alignments from the first n reads such that $n \% 100 = 1$ and the total alignment length reaches at least 1,000,000 bases. FMLRC2 corrected long reads have a mean length of 1,447 bases and 90.6 of them were aligned to the reference genome by Minimap2. The sample AlignQC would then use to evaluate the base error rates can therefore be expected to consist of only 801 long reads, and is not representative for the whole read set.

.3. rnaQUAST adaptation

Line 13 in `UtilsAnnotation.py`

```
default_type_isoforms = ['transcript', 'RNA', 'mRNA', 'miRNA', 'ncRNA', 'tRNA', 'rRNA']
```

defines the types of transcripts that will be used in the analysis. This list, however is incomplete for the GRCm39 annotation and I therefore added 'snRNA', 'lnc_RNA', 'snoRNA', 'tRNA', 'anti-sense_RNA', 'RNase_P_RNA', 'telomerase_RNA', 'RNase_MRP_RNA', 'Y_RNA' and 'scRNA' to this list. This adaptation raised the number of reported isoforms in the SQL-database used by rnaQUAST from 110,169 to 136,032.

I have also suggested this change to the authors of rnaQUAST, and it has already been accepted

.3. RNAQUAST ADAPTATION

(see Issue #27 in their github repository).

Nevertheless, the added transcript types have only slightly altered the results presented in this thesis. Some of the added transcript types are either not captured by commonly used RNASeq protocols or might only be present during certain phases of the cell cycle.

Statement of originality

I hereby confirm that I have written the accompanying thesis by myself, without contributions from any sources other than those cited in the text and acknowledgments. This applies also to all graphics, drawings, maps and images included in the thesis.

Jena, December 15, 2025

Place and date

Signature